

**CAMELLIA SINENSIS / ANANAS COMOSUS FIBER
BASED UNI-DIRECTIONAL CORRUGATED PANEL
FOR SOUND DAMPING APPLICATION**

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

In this environmental degrading world, noise pollution has its ripple effect on lives of human being. The interior control of the noise has been notable improving in order to save the future generation but at the same time the advance materials should also balance development and application where it is being used. Besides that, the material should not harm human beings at any cost. The use of renewable, cheap, and less risk natural fibers will lead to empirical waste management solutions. Due to higher weight of laminated composites now the people are fast emerging to corrugated structure which provides less weight and high performance. Camellia Sinensis / Ananas Comosus / GFRP / Epoxy Polymer composites are widely accepted to be a better material for sound damping materials. It also reduces the usage of the material which is highly dumped in laminated type. Two different natural fibers such as pineapple, tea powder 20wt. %, 10wt. % respectively are made in corrugated structure composites amalgamated by Epoxy matrix. The samples are subjected to acoustical testing and their output responses (sound absorbing capacity) are obtained. A best optimized solution is found and it can be chosen for the application of smart class room sound damping.

Keywords: Acoustic panel, corrugated structure, natural fiber, sounds absorption capacity.

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LIST OF SYMBOLS AND ABBREVIATIONS

GFRP	-	Glass Fiber Reinforced Polymer
μ	-	Micro
PALF	-	Pineapple Leaf Fiber
MMC	-	Metal Matrix Composites
CMC	-	Ceramic Matrix Composites
PMC	-	Polymer Matrix Composites
GRP	-	Glass-Reinforced Plastic
FRP	-	Fiber-Reinforced Polymer
OH	-	Hydroxide
NaOH	-	Sodium Hydroxide
FEM	-	Finite Element Modeling
CO ₂	-	Carbon Di Oxide
RNCF	-	Recycled Newspaper Cellulose Fiber
DPF	-	Date Palm Fiber
CCF	-	Coconut Coir Fiber
AAC	-	Acoustic Absorption Coefficients
BFRP	-	Bamboo Fiber Reinforced Polypropylene
PU	-	Polyurethane
PP	-	Polypropylene

EFB	-	Empty Fruit Bunch
MF	-	Mesocarp Fiber
SPL	-	Sound Pressure Level
OPEFB	-	oil palm empty fruit bunch
LC	-	Laminate composites
SEM	-	Scanning Electron Microscope
HLC	-	hybrid laminated composites
ASTM	-	American Standard
LED	-	Light Emitting Diode
PVA	-	Polyvinyl Alcohol
E_m	-	Modulus of Matrix
E_f	-	Modulus of Fiber
V_m	-	Volume Fractions of Matrix
V_f	-	Volume Fractions of Fiber
σ_f	-	Stresses in Fiber
σ_m	-	Stresses in Matrix
dB	-	Decibel
kHz	-	Kilohertz
NRC	-	Noise Reduction Coefficients

CHAPTER I

INTRODUCTION

The constituents of a composite are generally arranged in one or more discontinuous phases which are embedded in a continuous phase. The discontinuous phase is the reinforcement and the continuous phase is the matrix. In general, the reinforcement is much stronger and stiffer than matrix. Both constituents are required and each must accomplish specific tasks if the composite is to perform as intended. The physical and mechanical properties of composites depend on the properties, geometry, and concentration of the constituents. Increasing the volume content of reinforcements can increase the strength and stiffness of a composite. If the volume content of reinforcements is too high there will not be enough matrix to keep them separate, and they can become tangled. Similarly, the geometry of individual reinforcements and their arrangement within the matrix can affect designing with composite materials. The type of reinforcement and matrix, the geometric arrangement and volume fraction of each constituent, the anticipated mechanical loads, the operating environment for the composite must all be taken into account.

Based on the matrix material which forms the continuous phase, the composites are broadly classified into metal matrix (MMC), ceramic matrix (CMC) and polymer matrix (PMC) composites. Of these, polymer matrix composites are much easier to fabricate than MMC and CMC. This is due to the relatively low processing temperature required for fabricating polymer matrix composite.

Unlike conventional engineering materials, a composite material is generally non-homogeneous and does not behave as an isotropic material. Most composites behave as either anisotropic or orthotropic materials. The material properties of an anisotropic material are different in all directions. There is

typically a coupling of extension and shear deformation under conditions of uniaxial tension. There are varying degrees of anisotropic material behavior and the actual deformation resulting from applied loads depends on the material. The material properties of an orthotropic material are different in three mutually perpendicular planes, but there is generally no shear extension coupling as with an anisotropic material. The transverse in-plane and out-of-plane displacements are not typically same, because Poisson's ratio is different in these two directions. Although it appears similar to that of an isotropic material the magnitudes of the in-plane and out-plane displacement are different.

1.1 Classification of composite material

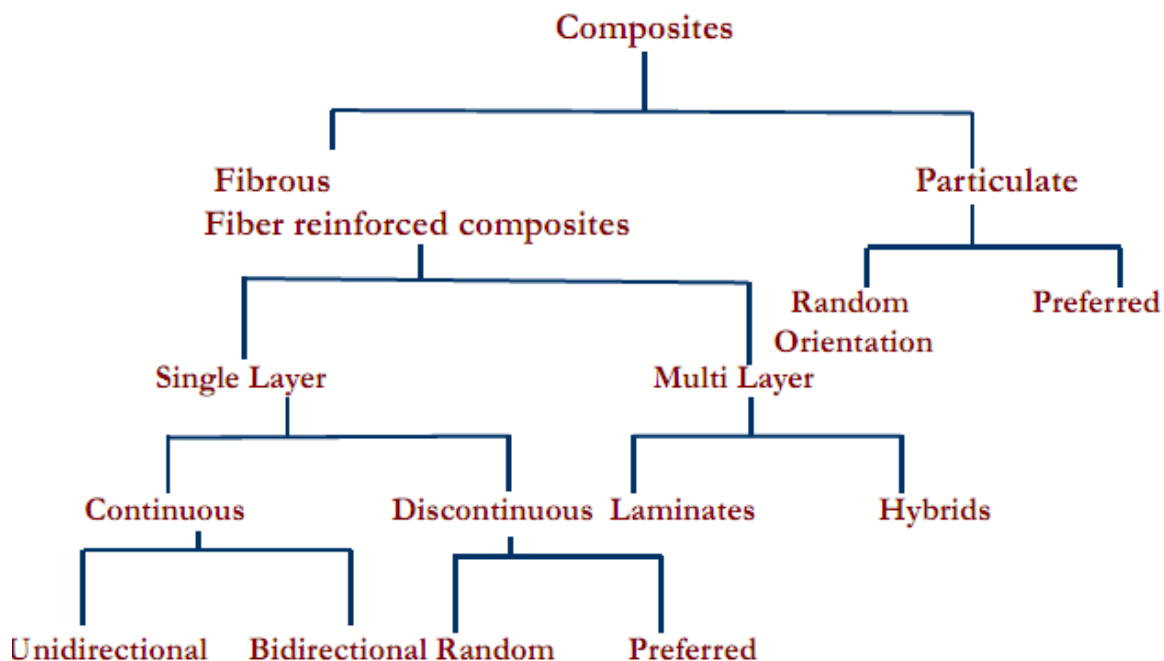


Fig 1.1 Classification of composite material

Fibrous: A fibrous composite consists either continuous (long) or chopped (whiskers) fibers suspended in a matrix material.

Continuous fibers: A Continuous fiber is geometrically characterized as having a very high length to diameter ratio. They are on the whole stronger and

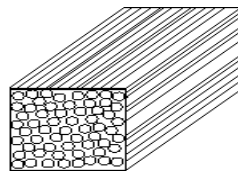
stiffer than bulk material. Depending upon the fiber, fiber diameters arrange between 3 and 200 μm in general.

Composite materials are usually classified according to the type of reinforcement used as shown in Fig 1.1

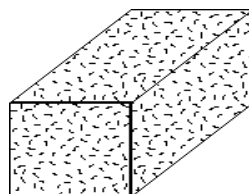
Whiskers: A whiskers is generally considered to be a short, stubby fiber. It can be broadly defined as having a length to diameter ratio of $5 < l/d < 1000$ and beyond. Whisker diameters generally arrange between 0.02 and 100 μm . composite in which the reinforcement are discontinuous fibers or whiskers can be produced.

Particulate composites: It consists of particles immersed in matrices such as alloy and ceramics shown in Fig 1.2. They are usually isotropic because the particles are added randomly. It has advantages such as improved strength, increased operating temperature, oxidation resistance etc.

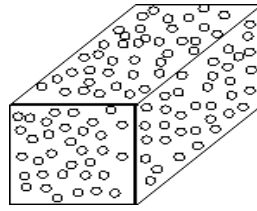
Example: Aluminum particles in rubber, silicon carbide particles in aluminum.



(a) Random fiber reinforced composites



(b) Continuous fiber reinforced composites



(c) Particulate reinforcement composites

Fig 1.2 Schematic representations of composites

Flake composites: It consist of flat reinforcements of matrices shown in Fig 1.3. Typical flake materials are glass, mica, aluminum and silver.

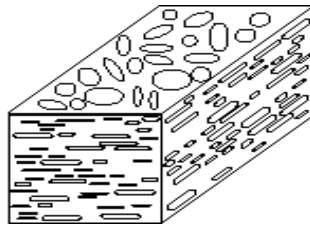


Fig 1.3 Schematic representations - flat flakes as the reinforcement

Fiber reinforced composites are usually used in the forms of laminates. A laminate is made by bonding two or more lamina together. Lamina usually is made of parallel fibers set in a polymer matrix at some constant volume fraction. The laminate is assembled with the individual lamina at different fiber orientations. Fig 1.4 shows the individual laminate (which are at different angular orientations) as well as a fully assembled FRP composite.

The load-response behavior and engineering properties of the laminate are determined by both the fiber orientations in the lamina and the order in which they are assembled together in the laminate.

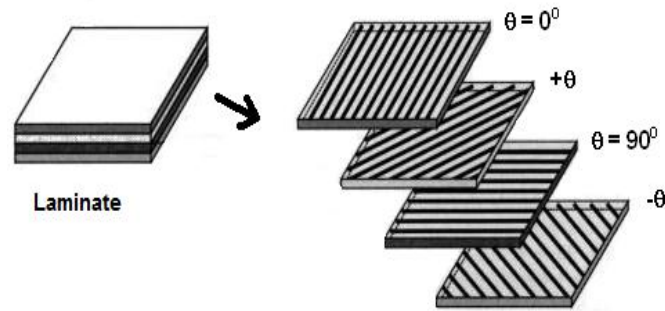


Fig 1.4 Laminated Fiber-reinforced Composite

1.2 Terminologies of composite material

Lamina: A lamina is a flat or a curved arrangement of unidirectional or woven fibers suspended in a matrix material. A lamina is generally assumed to be orthotropic, and the material from which it is made designate its thickness.

Fibers: Fibers are a special case of reinforcements. They are generally continuous and have diameters ranging from 3 to 200 μm . Fibers are typically linear elastic or elastic perfectly and are generally stronger and stiffer than the same materials in bulk form. The most commonly used fibers are boron, glass, carbon and kevlar.

Matrix: The matrix is the binder material that supports, separates and protects the fibers. It provides path by which load is both transferred to the fibers and redistributed among the fibers in the event of fibers breakage. The matrix typically has a lower density, stiffness, and strength than the fibers. Matrices can be brittle, ductile, elastic or plastic. They can have either linear or nonlinear stress-strain behavior. In addition the matrix material must be capable of being forced around the reinforcement during some stage in the manufacture of the composite. Fibers must often be chemically treated to ensure proper adhesion to the matrix. The most commonly used matrices are carbon, ceramic, metal and polymeric.

Carbon matrix composites: A carbon matrix has a high heat capacity per unit weight. They have been used as rocket nozzles, ablative shields for reentry vehicles, and clutch and brake pads for aircraft.

Ceramic matrix composites: A ceramic matrix is usually brittle. Carbon, ceramic, metal and glass fibers are typically used with ceramic matrices in areas where extreme environments are anticipated.

Glass matrix composites: Glass and glass ceramic composites usually have an elastic modulus much lower than that of the reinforcement. Carbon and metal oxide fibers are the most common reinforcements with glass matrix composites. The best characteristics of glass or ceramic matrix composites are their strength at high service temperatures. The primary applications of glass matrix composites are for heat resistant parts in engines, exhaust systems and electrical components.

Metal matrix composites: A metal is especially good for high-temperature use in oxidizing environments. The most commonly used metals are iron, nickel, tungsten, titanium, magnesium and aluminum. There are three classes of metal matrix composites.

Class 1. The reinforcement and matrix are insoluble. Reinforcement / matrix combinations in this class include tungsten or alumina / copper, BN - coated B or boron/aluminum etc.

Class 2. The reinforcement / matrix exhibits some solubility and the interaction will alter the physical properties of the composite. Reinforcement / matrix combinations included in this class are carbon or tungsten / nickel, tungsten / columbium and tungsten / copper.

Class 3. The most critical situations in terms of matrix and reinforcements are in this class. The problems encountered here are generally of a manufacturing

nature and can be solved through processing controls. Within this class the reinforcement / matrix combinations include alumina or boron or silicon carbide / titanium.

Polymer matrix composites: Polymeric matrices are the most common and least expensive. They are found in nature as amber, pitch and resin. Some of the earliest composites were layers of fiber, cloth and pitch. Polymers are easy to process, offer good adhesion. They are a low-density material. Because low processing temperatures, many organic reinforcements can be used. A typical polymeric matrix is either viscoelastic or viscoplastic, meaning it is affected by time, temperature, and moisture.

Thermoset: A thermoset matrix has highly cross - linked polymer chains. After it has been processed. A thermoset can not be remolded.

Thermoplastic: A thermoplastic matrix has polymer chains that are not cross – linked. Although the chains can be in contact, they are not linked to each other. A thermoplastic can be remolded to a new shape when it is heated to the same formations temperature.

Laminate: A lamina is made of two or more unidirectional lamina or plies stacked together at various orientations. The various laminate designations are listed in Table 1.1

Composite laminates containing plies of two or more different types of materials are called hybrid composites and more specifically interply hybrid composites. Examples: a composite laminate may be made up of unidirectional glass / epoxy, carbon / epoxy and aramid / epoxy layers stacked together in a specified sequence. In some cases it may be advantageous to inter mingle different types of fibers, such as glass and carbon or aramid and carbon, with in the same unidirectional ply. These composites are intraply hybrid composites.

One could combine intraply hybrid layer with other layers to form an intraply - interply hybrid composite.

Table 1.1 Designations of laminates

Unidirectional 6 – ply	$[0/0/0/0/0/0] = [0_6]$
Cross ply	$[0/90/90/0] = [0/90]_s$
Angle – ply symmetric	$[+45/-45/-45/+45] = [\pm 45]_s$ $[30/-30/30/-30/-30/30/-30/30] = [\pm 30]_{2s}$
Angle – ply asymmetric	$[30/-30/30/-30/30/-30/30/-30] = [\pm 30]_4$
Multi directional	$[0/45/-45/-45/+45/0] = [\pm 45]_s$ $[0/0/45/-45/0/0/0/0/-45/45/0/0]_s = [0_2/\pm 45/0_2]_s$ $[0/15/-15/15/-15/0] = [0/(\pm 15)_2/0]_s$

** S = Symmetric sequence

1.3 Characterization of composite material

Fibers should have a following requirements to fullfil their reinforcement property and to use these in applications such as ropes, woven fabrics and clothes.

Requirements:

1. Lateral dimension in millimeter or micrometer in size.
2. Flexible
3. Fine fibers have small flaws and defects there by increasing the strength.
4. Fiber must be several times stronger than the matrix to share the higher fraction of the total load.

5. Higher elastic properties than the matrix

1.4 Classification of fibers

Fibers are a class of hair-like material that are continuous filaments or are in discrete elongated pieces, similar to pieces of thread. They can be spun into filaments, thread, or rope. They can be used as a component of composites materials. They can also be matted into sheets to make products such as paper or felt. Fibers are of two types: natural fiber and manmade or synthetic fiber. Fig 1.5 shows the classification of natural fibers.

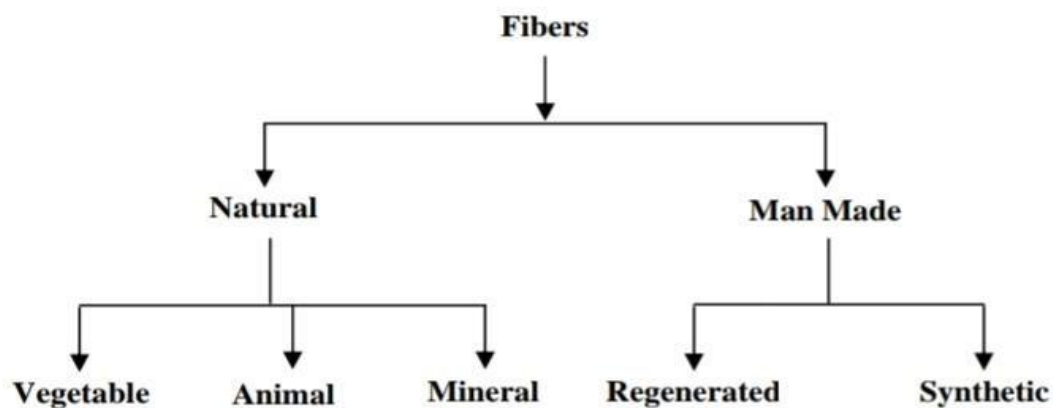


Fig. 1.5 Classification of fibers

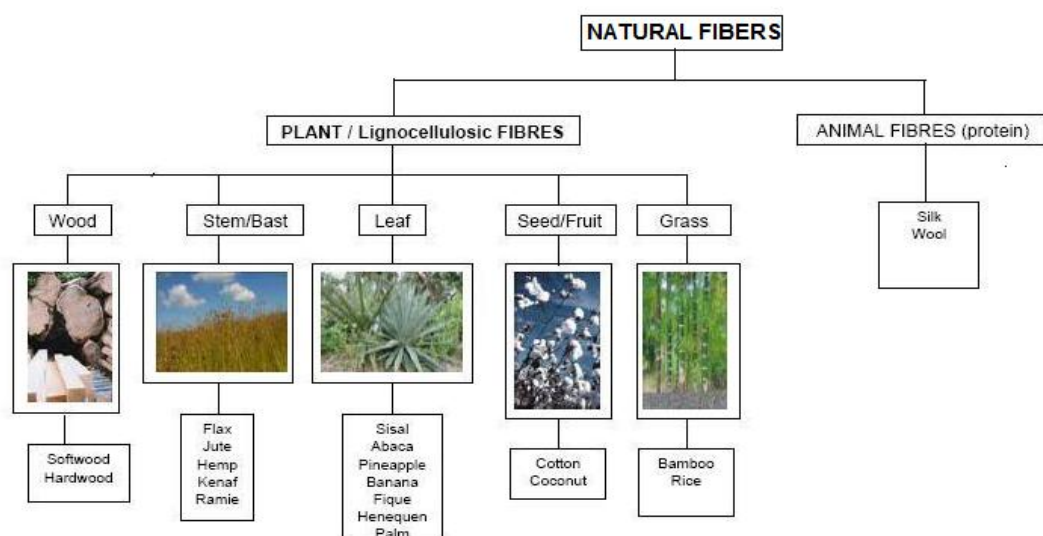


Fig 1.6 Classification of natural fibers

1.5 Sources of Natural Fibers

Natural fibers include those made from plant, animal and mineral sources. Natural fibers can be classified according to their origin

1.5.1 Vegetable fibers

Vegetable fibers are generally comprised mainly of cellulose: examples include pineapple, cotton, jute, flax, ramie, sisal, and hemp. Cellulose fibers serve in the manufacture of paper and cloth. This fiber can be further categorized into the following:

- Seed fiber: Fibers collected from seeds or seed cases. E.g. cotton and kapok.
- Leaf fiber: Fibers collected from leaves. E.g. sisal and agave.
- Blast fiber or skin fiber: Fibers are collected from the skin or baste surrounding the stem of their respective plant. These fibers have higher tensile strength than other fibers. Therefore, these fibers are used for durable yarn, fabric, packaging, and paper. Some examples are flax, jute, kenaf, industrial hemp, ramie, rattan, soybean fiber, and even vine fibers and banana fibers.
- Fruit fiber: Fibers are collected from the fruit of the plant, e.g. coconut (coir) fiber.
- Stalk fiber: Fibers are actually the stalks of the plant. E.g. straws of wheat, rice, barley, and other crops including bamboo and grass. Tree wood is also such a fiber.

The most used natural fibers are cotton, flax and hemp, although sisal, jute, kenaf, and coconut are also widely used. Hemp fibers are mainly used for ropes and aero foils because of their high suppleness and resistance within an aggressive environment. Hemp fibers are, for example, currently used as a seal within the heating and sanitary industries.

(a) Pineapple fiber

Natural fiber based composites are under intensive study due to their ecofriendly nature and peculiar properties. The advantage of natural fibers is their continuous supply, easy and safe handling, and biodegradable nature. Although natural fibers exhibit admirable physical and mechanical properties, it varies with the plant source, species, geography, and so forth. Pineapple leave fiber (PALF) is one of the abundantly available wastes materials of Malaysia and has not been studied yet as it is required. A detailed study of chemical, physical, and mechanical properties will bring out logical and reasonable utilization of PALF for various applications. From the socioeconomic prospective, PALF can be a new source of raw material to the industries and can be potential replacement of the expensive and nonrenewable synthetic fibre. Natural tea-leaf fibers were used as reinforcement agents in different polyurethane-based matrixes to improve the sound absorption properties of the composite. Adding tea-leaf-fibers to rigid polyurethane foam demonstrate limited contribution to sound absorption properties of the material. Furthermore, adding tea-leaf-fibers to soft foam results in a significant improvement of sound absorption properties of the composite.

(b) Tea fiber

Natural tea-leaf fibers were used as reinforcement agents in different polyurethane-based matrixes to improve the sound absorption properties of the composite. Adding tea-leaf-fibers to rigid polyurethane foam demonstrate limited contribution to sound absorption properties of the material. Furthermore, adding tea-leaf-fibers to soft foam results in a significant improvement of sound absorption properties of the composite. Natural tea-leaf fibers were used as reinforcement agents in different polyurethane-based matrixes to improve the sound absorption properties of the composite. Adding tea-leaf-fibers to rigid polyurethane foam demonstrate limited contribution to sound absorption

properties of the material. Furthermore, adding tea-leaf-fibers to soft foam results in a significant improvement of sound absorption properties of the composite.

1.5.2 Animal fibers

Animal fibers generally comprise proteins; examples include silk, wool, angora, mohair and alpaca.

- Animal hair (wool or hairs): Fiber or wool taken from animals or hairy mammals. E.g. sheep's wool, goat hair (cashmere, mohair), alpaca hair, horse hair, etc.
- Silk fiber: Fiber collected from dried saliva of bugs or insects during the preparation of cocoons. Examples include silk from silk worms.
- Avian fiber: Fibers from birds, e.g. feathers and feather fiber.

1.5.3 Mineral fibers

Mineral fibers are naturally occurring fiber or slightly modified fiber procured from minerals. These can be categorized into the following categories:

- Asbestos: The only naturally occurring mineral fiber. Variations are serpentine (chrysotile) and amphiboles (amosite, crocidolite, tremolite, actinolite, and anthophyllite).
- Ceramic fibers: Glass fibers (Glass wool and Quartz), aluminum oxide, silicon carbide, and boron carbide.
- Metal fibers: Aluminum fibers.

1.5.4 Synthetic fibers

Synthetic fibers such as glass fibers, carbon fibers, and boron fiber are gaining more attention over last two decades. Synthetic fibers are made from raw materials such as petroleum, based on chemicals or petrochemicals. These materials are polymerized into a long, linear chemical with different chemical compounds and are used to produce various types of fibers. Synthetic fibers

account for about half of all fiber usage, with applications in every field of fiber and textile technology

(a) Glass fiber

Glass fiber is the best-known reinforcement in high performance composite applications due to its appealing combination of good properties and low cost. The major ingredient of glass fiber is silica which is mixed with varying amounts of other oxides. The different types of glass fibers commercially available are E and S glass. The letter 'E' stands for 'electrical' as the composition has a high electrical resistance and 'S' stands for strength.

Glass fibers are used successfully for reinforcing the plastics and therefore, the suitability of this fiber as a reinforcing material for rubbers has been studied. High initial aspect ratio can be obtained with glass fibers, but brittleness causes breakage of fibers during processing

1.6 Structural compositions of natural fiber

All the natural fibers having natural constituents like cellulose, hemicellulose, lignin, pectin and wax content. The Cellulose micro fibrils have their own cell geometry which is a factor responsible for the properties of the fiber. Each fiber cell wall consists of primary and secondary layers of cellulose micro fibrils. The fiber structure develops in the primary cell wall and is deposited during its growth. The secondary wall consists of three layers and each layer has a long chain of helical cellulose micro fibrils. The cellulose content increases steadily from primary to secondary layers and the hemicelluloses amount are similar in each layer. However, lignin content decreases in this sequence. Hemicellulose molecules are hydrogen bonded with cellulose fibrils and they form cementing materials for the fiber structure. Lignin and pectin are coupled with the cellulose– hemicellulose network and provides an adhesive quality to hold the molecules together is listed in Table 1.3. This adhesive quality is the cause for the strength and stiffness properties of

the fiber. Secondary thick layer determines the mechanical properties of the fiber. Generally, fibers with higher cellulose content and a lower microfibrillar angle (the angle between the fiber axis and cellulose micro fibrils) have better strength properties. Although natural fibers have relatively lower strength properties compared to the synthetic fibers, the specific modulus and elongation at break signifies the potentiality of these fibers to replace synthetic fibers in engineering polymer composites.

1.6.1 Matrix

A matrix is a binder material that is used to hold fibers in position and transfer external loads to internal reinforcements. In natural fiber reinforced polymer composites, both thermoset and thermoplastic matrices such as unsaturated polyesters, epoxies and phenolic, and polypropylenes, polyethylene and elastomers, respectively are widely used for composites applications. These matrices have different chemical structures and undergo different reactivity's with the surface molecules of fibers in composites. Table 1.4 represents the comparison of thermoset and thermoplastic matrices.

1.6.2 Characteristics of good resin

1. Good impregnation to allow all the different fibres to bind with resin to form a single material.
2. Transfers load to the fibre through the fibre-matrix interface.
3. Protects fibres from environmental damage.
4. Non-toxic, since major applications are domestic it should not create any harmful effects due to change in environmental conditions.
5. High availability, it should be readily available at affordable cost.

Table 1.2 Comparison between thermoplastic and thermoset matrix

Advantages	Disadvantages
Thermoset - Low Resin viscosity - Good fiber wetting - Excellent thermal stability - chemically resistant Thermoplastic - Recyclable - Easy to repair by welding - Post formable - Tough	- Brittle - Non-recyclable via - Standard techniques - Not post-formable - Poor melt flow - Need to be heated above the melting point for processing purposes

1.7 surface modification of natural fiber

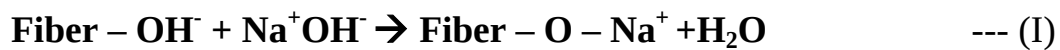
1.7.1 Chemical methods

The major problem of natural fiber composites originates from the hydrophilic nature of the fiber and hydrophobic nature of the matrix. The inherent incompatibility between these two phases results weakening bonding at the interface. Chemical treatments on reinforcing fiber can reduce its hydrophilic tendency and thus improve compatibility with the matrix. Several research activities have been conducted to improve fiber adhesion properties with the matrix through chemical treatments. The followings are the reviews of

different chemical treatments on the fiber and their effects on composite properties.

1.7.2 Alkaline Treatment

Alkaline treatment or mercerization is one of the most used chemical treatments of natural fibers when used to reinforce thermoplastics and thermosets. The important modification done by alkaline treatment is the disruption of hydrogen bonding in the network structure, thereby increasing surface roughness. This treatment removes a certain amount of lignin, wax and oils covering the external surface of the fiber cell wall, depolymerizes cellulose and exposes the short length crystallites. Addition of aqueous sodium hydroxide (NaOH) to natural fiber promotes the ionization of the hydroxyl group to the alkoxide as shown in equation (I).



Thus, alkaline processing directly influences the cellulosic fibril, the degree of polymerization and the extraction of lignin and hemi cellulosic compounds. It is reported that alkaline treatment has two effects on the fiber:

- 1) It increases surface roughness resulting in better mechanical interlocking, and
- 2) It increases the amount of cellulose exposed on the fiber surface, thus increasing the number of possible reaction sites.

1.8 Corrugated structure

In mechanical engineering any structure which has a surface with the shape of corrugation either made by folding, molding, or any other manufacturing methods is called a corrugated structure.

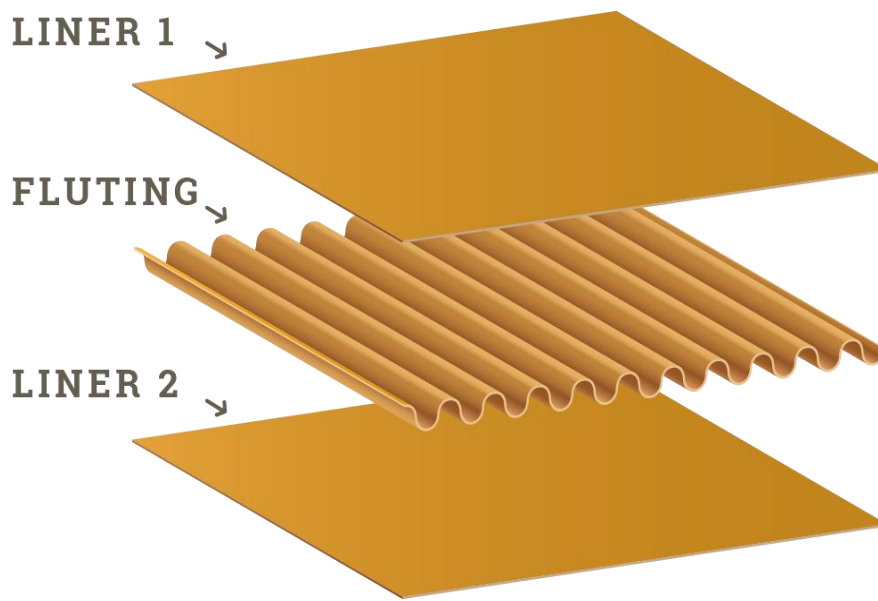


Fig.1.7 Corrugated structure

Corrugation has long been seen as a simple and effective means of forming lightweight structures with high anisotropic behavior, stability under buckling load and energy absorption capability. This has been exploited in diverse industrial applications and academic research. In recent years, there have been numerous innovative developments to corrugated structures, involving more elaborate and ingenious corrugation geometries and combination of corrugations with advanced materials. This development has been largely led by the research interest in morphing structures, which seek to exploit the extreme anisotropy of a corrugated panel, using the flexible degrees of freedom to allow a structure's shape to change, whilst bearing load in other degrees of freedom.

A comprehensive review of the literature on corrugated structures, with applications ranging from traditional engineering structures such as corrugated steel beams through to morphing aircraft wing structures. As such it provides an important reference for researchers to have a broad but succinct perception of

the mechanical behavior of these structures. Such a perception is highly required in the multidisciplinary design of corrugated structures for the application in morphing aircraft.

1.8.1 Types of corrugated structure

Corrugated plates are assumed to have trapezoidal or sinusoidal corrugation in one direction as shown in Figures 1 and 2. The corrugated plate can be described by the number of corrugations N_c , the pitch C , the height F , and the trough angle θ (for trapezoidal corrugation). In this work, FEM is used for the modal analysis of corrugated plates. In addition to FEM, the corrugated plates are studied analytically based on modeling the corrugated plates as orthotropic flat plates with the same thickness and equivalent rigidities.

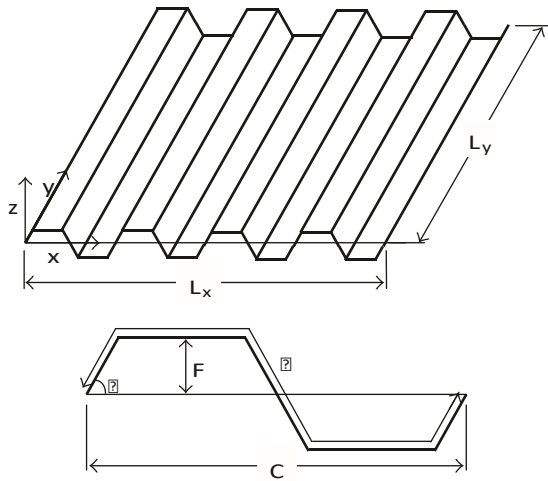


Fig 1.8 Trapezoidal corrugated plate.

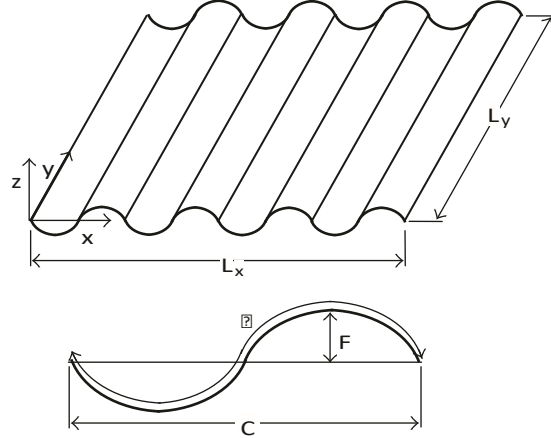


Fig 1.9 Sinusoidal corrugated plate.

1.8.2 Merits of corrugated structure

The main advantages of corrugated structure composite are,

- The corrugated structure is mainly preferred for less weight application.
- Comparing to normal acoustic panel, corrugated structured panel gives equivalent sound absorption with less weight.

- The corrugated structure has different types such as honeycomb, sine wave, curved and etc.
- Their repetitive folds on the surface are what offer them the better durability and enhanced strength that they possess.
- They also pave the way for increased strength across smaller surface areas.
- Corrugated sheets can also withstand strong impacts from objects falling onto them.
- Because of their exceedingly anisotropic behavior of these structures they have been proposed as a flexible skin for the wings of morphing aircraft.
- Their multifunctional integral thermal protection structures protect the spacecraft from extreme reentry temperatures, and possess load-carrying capabilities

1.9 Merits of natural fiber composite

The main advantages of natural fiber composite are,

- Low specific weight, resulting in a higher specific strength and stiffness than glass fiber.
- It is a renewable source, the production requires little energy, and CO₂ is used while oxygen is given back to the environment.
- Producing with low investment at low cost, which makes the material an interesting product for low wage countries.
- Reduced wear of tooling, healthier working condition, and no skin irritation.
- Thermal recycling is possible while glass causes problem in combustion furnaces.
- Good thermal and acoustic insulating properties.

1.10 Applications of Natural Fiber Composites

The natural fiber composites can be very cost-effective material for following applications such as,

- Building and construction industry: panels for partition and false ceiling, partition boards, wall, floor, window and door frames, roof tiles, mobile or pre-fabricated buildings which can be used in times of natural calamities such as floods, cyclones, earthquakes, etc.
- Storage devices: post-boxes, grain storage silos, bio-gas containers, etc.
- Furniture: chair, table, shower, bath units, etc.
- Electric devices: electrical appliances, pipes, etc.
- Everyday applications: lampshades, suitcases, helmets, etc.
- Transportation: automobile and railway coach interior, boat, etc.
- The reasons for the application of natural fibers in the automotive industry include:
 - Low density: which may lead to a weight reduction of 10 to 30%
 - Acceptable mechanical properties, good acoustic properties.
 - Favorable processing properties, for instance low wear on tools, etc.
 - Options for new production technologies and materials.
 - Favorable accident performance, high stability, less splintering.
 - Favorable Eco balance for part production.
 - Favorable Eco balance during vehicle operation due to weight savings.
 - Occupational health benefits compared to glass fibers during production. No off-gassing of toxic compounds (in contrast to phenol resin bonded wood and recycled Cotton fiber parts).
 - Reduced fogging behavior.
 - Price advantages both for the fibers and the applied technologies.

CHAPTER II

LITERATURE REVIEW

Samson Rwawiire et al. (2015) described the quest for sound-absorbing materials that are not only environmentally friendly, but also sustainable is the foremost reason for natural fiber-acoustic materials. Bark cloth is a natural non-woven fabric that is largely produced from *Ficus* trees. An exploratory investigation of bark cloth a non-woven material and its reinforcement in epoxy polymer composites has been fabricated and investigated for the sound absorption properties so as to find the most suitable applications and also to see whether bark cloth can be used in some applications in place of man-made fibers. Three types of material species were investigated with their respective composites. The fiber morphology showed bark cloth to be a porous fabric that showed promising sound absorption properties at higher frequencies. The sound absorption results of four-layer material selections of *Ficus natalensis*, *Ficus brachypoda* and *Antiaris toxicaria* bark cloth showed sound absorption coefficient of 0.7; 0.71 and 0.91 at $f > 6400$ Hz, respectively. The bark cloth reinforced laminar epoxy composites had reduced sound absorption coefficients, which ranged from 0.1 to 0.35, which was attributed to decreased porosity and vibration in the bark cloth fiber network.

Rao et al. (2014) aims to introduce new natural fiber, used as fillers in a polymeric enabling production of economical and lightweight composites for load carrying structures. An investigation of the extraction procedure of vakka, date and bamboo has been undertaken. The cross-sectional shape, the density and tensile properties of these fibers, along with established fibers like sisal, banana, coconut and palm, are determined experimentally under similar condition and compared. The fibers introduced in the present study could be

used as an effective reinforcement for making composites, which have an added advantage of being lightweight.

Satish et al. (2015) conducted their study on their effects of hybrid composites specimen subjected to in plane tensile and compressing loading. He concluded that the increase in shear stress was not only dependent on fiber strength but also on the interface between the fiber and matrix material. Besides, the mechanical properties of hybrid composites were investigated by and result showed that increasing the volume fraction of matrix could enhance the energy absorbing characteristics of the hybrid composites were increasing the amount of fiber increased the flexural modulus.

Ray et al. (2016) have studied the effects of alkali treatment on the mechanical properties of sisal composites. The result showed that composites which received from alkali treatment had better mechanical properties this may be due to fact that the treatment includes the adhesive characteristics of fiber surface and improves the impact strength of the composites.

Huda et al. (2012) evaluated the effect of the addition of silane-palc as fillers on the mechanical and physic-mechanical properties of poly recycled newspaper cellulose fiber (RNCF)/ talc hybrid composites; the hybrid composites showed improve properties such ad flexural strength and modulus the mechanical properties of banana/kenaf hybrid composites subjected to various chemical treatment such ad sodium lauryl sulfate (SLS) and sodium hydroxide

Ancuța-Elena Tiuc et al. (2015) narrated that the Polyurethane foam is an outstanding material for various applications. It is manufactured by propelling liquid isocyanate-polyol mixture to form foams in the presence of a blowing agent. This paper comprises an experimental study on acoustic properties improvement of rigid polyurethane closed-cell foam, by

incorporating various quantities of textile waste into the matrix. In order to obtain a homogenous, easy to handle material, an optimal percent of 10-50% textile waste was used. The sound absorption coefficient of the composite materials was measured using an impedance tube. The composite materials obtained have better sound absorption properties compared to rigid polyurethane foam. The noise reduction coefficient (NRC) of the composite material with 40% textile waste and 60% rigid polyurethane foam is twice as high as the 100% rigid polyurethane material. Properties of materials obtained in this study are of a great importance for improving the existing materials properties and for developing new materials with better physical, mechanical and thermal properties. These alternative materials can be cost beneficial as well as useful in green building initiative by developing materials from natural and recycled resources. Composite materials obtained by introducing textile waste in rigid polyurethane foam matrix have sound absorption properties much improved compared to the 100% rigid polyurethane foam. The 60-RPF material has the best capacity to absorb noise, reaching a value of minimum 0.4 of sound absorption coefficient on the whole analyzed frequency ranges. The developed materials can be used both in the outdoor and in the indoor environment.

Cicala et al. (2016) have studied the properties and performance of various hybrid glass/natural fiber composites for the applications in curved pipes. Natural fibers are lighter and cheaper, but they have low mechanical properties than glass fibers. The use of hybrid fibers may solve this issue. Most of the studies on natural fibers are concerned with single reinforcement. The addition of natural fiber to the glass fiber can make the composite hybrid which is comparatively cheaper and easy to use.

Thwe and Liao (2015) have studied the durability of bamboo/glass fiber reinforced polymer matrix hybrid composites. They have studied the properties such as tensile strength and elastic modulus of bamboo fiber reinforced

polypropylene (BFRP) and bamboo glass fiber reinforced polypropylene hybrid composites. They have indicated that the tensile strength and elastic modulus decreased after ageing.

L.A. ALRahman et al (2013) discussed that the natural fiber materials are one of the major ways to control environmental pollution and new materials need to be considered to find the best solutions. This study presents an experimental investigation on pure micro porous materials from two types of fibers: Date Palm Fiber (DPF) and Coconut Coir Fiber (CCF). This study was conducted to examine the potential for using these two types of fibers as sound absorbers. To account for the effects of the characteristics of these materials, such as thickness and density, on the acoustic absorption coefficient of a sound absorber, the measurements were conducted in an impedance tube on normal incidence acoustic absorption. The Acoustic Absorption Coefficients (AAC) of the two types of materials was increased at all frequencies when the thickness of the sample was increased. The results show that date palm fiber and coconut coir fiber have good acoustic properties at low and high frequencies and can be used as an alternative replacement to conventional products. The comparisons between the two panels show a good potential because they are cheaper and lighter in comparison to asbestos and rock wool industrial materials.

Jorge P. Arenas(2010) explained that the Sound-absorbing materials absorb most of the sound energy striking them, making them very useful for the control of noise. They are used in a variety of locations – close to sources of noise, in various paths, and sometimes close to receivers. Although all materials absorb some incident sound, the term “acoustical material” has been primarily applied to those materials that have been produced for the specific purpose of providing high values of absorption. The major uses of absorbing materials are almost invariably found to include the reduction of reverberant sound pressure levels and, consequently, the reduction of the reverberation time in enclosures,

or rooms. A wide range of sound-absorbing materials exist. In the 1970s, public health concerns helped change the main constituents of sound-absorbing materials from asbestos-based materials to new synthetic fibers. Although, these new fibers are much safer for human health, more recently, issues related to global warming may increase the use of natural fibers instead of synthetic ones.

Y. Cao et al. (2006) studied the improvisation of mechanical properties of natural fiber in preparation of composites, heat and alkali treatments to kenaf fibers were performed and effects on mechanical properties were investigated. Compared with other heat treatment temperatures, the tensile strength of kenaf fiber at 140 °C exhibited maximum value, which could be attributed to the increase of crystallinity index of fibers after heat treatment. The result showed the fracture strain of the alkali treated fibers improved. Alkali treatment caused the elimination of hemicellulose and some impurities, and then led to the decreased crystallinity and the loosely bound structure of the fiber. This results in higher elongation at break of the treated fiber.

Silva et al. (2015) have developed the natural fibers/castor oil polyurethane composites and tested the fracture toughness. They have achieved the best fracture toughness performance for sisal fiber composites. The thermo physical properties of natural fiber reinforced polyester composites is carried out by and indicated that the natural fiber with glass allows a significantly better heat transport ability for the composites.

YANG WeiDong et al. (2012), reported about Sound absorption coefficients of three types of natural fibers, i.e., ramie, flax and jute fibers and their composites were measured by the two-microphone transfer function technique in the impedance tube. The results were compared with synthetic fibers and their composites. It was found that both natural fibers and their composites had superior capability of noise reduction. The multi-scale and

hollow lumen structures of natural fibers contributed to the high sound absorption performance. Moreover, the sound absorption properties of these natural fibers were also calculated by the Delany-Bazley and Garai-Pompoli models. They showed good agreement with the experimental data. It was concluded that multi-functional composite materials can be made by natural fibers so that both the mechanical and acoustical functions can be achieved. The sound absorption property of natural fibers was superior to synthetic fibers such as glass and carbon fibers due to their unique hollow and multi scale structures. Natural fiber reinforced composites also possessed better acoustic absorption behavior than synthetic fiber reinforced composite, especially at high frequencies, which might be very beneficial for the aeronautical applications. The theoretical predication values obtained from Garai-Pompoli model and Delany-Bazley model showed good agreement with the experimental results, which implied that the sound absorption coefficient of natural fibers and their composites can be predicted by these two models.

Hasina Mamtaz et al, (2016) explained the Natural fibers have already confirmed their potentiality in replacing common synthetic fibrous materials for acoustic absorption purposes. The study highlighted the possible ways to improve the acoustic behavior of natural fiber composites as high-quality absorbers, in combination with bio based granular materials. The alkaline treatment process causes the reduction of fiber diameter at the cost of the removal of moisture absorbents such as oil, cellulose, and wax of the natural fiber, thus improving adhesion and antifungus qualities of the composite. However, limited research was reported on the effect of the sound absorption performance due to alkaline treatment. More research is needed on the effect of the mercerization or alkaline treatment for acoustic absorption performance. In fibre granular composite materials, there is good potential in filling the small pores by the granular component and the formation of bridges between the

fibers as well. This contributes a higher surface area within the composite. Materials with higher density show increased absorption performance, since the density has a great influence on the porosity and flow resistivity of the composite. The acoustic absorption of natural fiber composites can be estimated by using the Delany-Bazley, Biot-Allard, and Johnson-Champoux-Allard analytical models. The Delany Bazley model is the only method that shows the general absorption pattern at overall broadband frequency without giving any information on the peaks and resonance of the frame. The other two models give accurate information about peaks and resonance of the frame.

Nasmi Herlina Sari et al. (2016) explained this research focuses on the development of a sustainable acoustic material comprising natural fibers of corn husk that were alkali modified by 1%, 2%, 5%, and 8% NaOH. The morphology and the acoustical, physical, and mechanical properties of the resulting fibers were experimentally investigated. Five different types of sample were produced in panel form, the acoustical properties of which were studied using a two-microphone impedance tube test. The sound absorption and tensile properties of the treated panels were better than those of raw fiber panels; the treated panels were of high airflow resistivity and had low porosity. Scanning electron micrographs of the surfaces of the corn husk fibers revealed that the different sound absorption properties of these panels were due to roughness and the lumen structures. The use of corn husk fiber for sound absorption was reported herein. The sound absorption approached 100% in the low-frequency range of 1.6–3.250 kHz for fiber samples that were treated with 2% and 5% NaOH concentrates. Panels of treated fibers were better in absorbing sound than the panels of raw fibers. In addition, the alkali treatment helped to maintain the panel shape and protect it the panel shape of fiber damage.

Venkateshwaran et al. (2014) studied the mechanical and water absorption properties of banana/sisal reinforced hybrid composites for taking

the length of the fiber and weight percentage as main constituent. They have reported the hybridization of sisal fiber with banana/epoxy composites up to 50% by weight increasing the mechanical properties and decreasing the water absorption properties. The overall tensile and flexural properties of natural fiber reinforced polymer hybrid composites are highly dependent on the aspect ratio, moisture absorption tendency, morphology and dimensional stability of the fibers used. The tensile and flexural properties of chemically treated natural fiber composites are slightly improved than the non-treated composites.

Ramesh et al. (2015) evaluated the properties of hybrid glass fiber – sisal/jute reinforced epoxy composites. The result indicated that the incorporation of sisal fiber with GFRP exhibited superior properties than the jute fiber reinforced GFRP composite in tensile properties and jute fiber reinforced GFRP composites performed better in flexural properties.

Ridzuan et al. (2012) analyzed the thermal and dynamic mechanical analysis of pennisetum purpureum GFRP epoxy hybrid composites. The fibers are treated by a process of mercerization (alkalization) and fabricated using 5% and 10% alkali treated fibers. The volume fractions of composite are 24% of fiber, 6% of glass fiber and 70% of epoxy resin. At $>60^{\circ}\text{C}$ debonding of fiber and matrix occurred, which resulted in reduction of the tensile and flexural strengths. The 5% alkaline treated fiber material interfacial shows a good surface morphology.

A.Rasdan Ismail, (2009) discussed the sound absorption capacity of natural coir fiber. Absorption coefficient of coir fiber is calculated using Delany-Bazley model. Experiments are conducted in impedance tube to validate the analytical outcomes for three samples having different thickness. Increasing the thickness of coir fiber enhances sound absorption and shifts noise absorption peak towards low frequency. Moreover, effect of air gap on sound absorption

performance of coir fiber has also been investigated. Increasing air gap thickness usually shifts the peak of absorption coefficient towards lower frequency. Then, the noise absorption of coir fibers with and without air gap is compared and shows that earlier one has greater absorption at low frequency than the later one. However, resonances cannot be predicted using this technique. Results also indicate that Delany-Bazley equations can justify the overall trend of absorption coefficient of coir fiber perfectly as demonstrated in this study.

Sushanta K.S et al. (2015) studied polypropylene–sisal/glass fiber hybrid composites mechanical analysis and fabrication as well as its dynamic behavior. Moreover, hybrid composites of polypropylene reinforced with sisal and glass fibers (SGRP) were fabricated using an intermeshing counter rotating twin screw extruder followed by injection molding. Maleic anhydride grafted polypropylene (MAPP) has been used as a coupling agent to improve the interfacial interaction between the fibers and matrix. The mechanical properties of the hybrid composites were studied from tensile, flexural, and impact tests.

Panthapulakkal and Sain, (2019) studied the mechanical and thermal properties of sisal/glass fiber–polypropylene (PP) composite materials. They have observed that the use of hybrid composite material enhances the flexural and impact properties. In addition, they have observed that the addition of glass fiber into sisal–PP composites resulted in improved thermal properties as well as the water resistance of the composites.

S. Gokulkumar et al. (2019) discussed on fiber-reinforced composite materials are made with *Camellia sinensis*–*Ananas comosus* hybridized by glass fiber and epoxy resin matrix. The amalgamated laminates were forged by a compression molding process with the composition of 40% wt. of reinforcing fiber and 60% wt. of epoxy matrix in a constant proportion. A hydrophilic

content present inside natural fibers are alkalized by 5% of NaOH to modify as hydrophobic nature for intensifying bonding between fiber and matrix. These hybridized fibers are mechanically, chemically, morphologically, and acoustically analyzed to meet the requirements of mechanical and sound proofing properties. The results shows that 25% wt. fraction of *Camellia sinensis* amalgamate with unidirectional fiber shows better average acoustical absorption coefficient (AAC) of 0.46 at 2000–6300 Hz frequency range and compared with existing material to establish the potential of using it in sound-proofing acoustical applications such as architectural acoustics, sound recording and reproduction room, noise barrier walls, loudspeaker design, etc.

Jia-Horng Lin et al. (2015) explained about the mesh fabric and porous filler are combined with polyurethane (PU) foam to fabricate sound absorbing composite boards with puncture resistance and cushion protections, after which sound absorption and static puncture resistance are improved. Effects of fillers (vermiculites and perlites) content and mesh opening ratio on sound absorbing, puncture resisting and cushioning properties are explored respectively. Result shows that, mesh fabric improves the low-frequency coefficients, and also increases the puncture resistance by 2 times and 1.5 times for 2-mm-diameter and 4.5-mm-diameter probes penetrations. Absorption peak shifts to the lower frequency with increase of fillers content and mesh opening, but static puncture resistance against 4.5-mm-diameter probe decreases with mesh opening ratio. The fabricated composite foam boards can absorb 90 % of impact energy when compounding with 10 wt% vermiculite fillers and 45 % opening fabric, which shows the favorable cushioning protection. This mesh-reinforced composite foam boards can be applied as protective compartment walls whatever in gerocomium, women and children care center, kindergarten or karaoke TV room.

Mathan Sambu et al. (2012) researchers divert their attentions to find sustainable and eco-friendly materials such as natural fibers to be an alternative sound absorber. This paper discusses the use of natural fiber from the Kenaf fiber (*Hibiscus Cannabinus*), that eligible as sound absorber. Sound absorption properties of this natural fiber have been investigated. The thickness of each sample is kept constant at 50mm only. To obtain the sound absorption, the Kenaf fiber samples was measured using impedance tube. The measured values from impedance tube were compared with the empirical model Delany- Bazley in order to investigate sound absorption's prediction result. In addition, bulk density, porosity and flow resistivity were also been examined. The result shows that Kenaf reach the maximum absorption coefficient value towards low frequency with fiber samples bind with natural rubber whereas fibers without the binder reach at high frequency range of 4000 Hz and above.

N I Khasanah, (2019) explained about absorption coefficient of nonwoven pineapple leaf fiber (PNW), and its layered combination with double weave (PDW) and tricot knitting fabric (PTK). Absorption coefficient test results showed there was an effect in absorption coefficient on nonwoven with layered. Cavities in the fiber are able to absorb sound at low and high frequencies. Thickness and weight in square affect the fabric density which can increase sound absorption. In this study, nonwoven layered with double weave fabric and tricot fabric is increases sound absorption at high frequencies. Nonwoven layered using double weave fabric and tricot fabric can be tried to determine the effect on sound absorption and become a solution to add aesthetics.

S. Gokulkumar et al. (2019) explained development of a new class of material using eco-friendly fibers extracted from the environmental and industrial wastage which is to be better alternative for existing green composites. In this work, an industrial tea waste has been considered as filler

materials due to its excellent sound absorbing properties. In this context, three combinations of three different natural fibers (pineapple, areca and ramie) are hybridized with waste tea (powder form) and GFRP were fabricated by a compression molding process with the combinations of 40% reinforcement and 60% epoxy matrix material in a constant proportion. The results are evaluated and compared to establish the possibility of using it as a new material for engineering applications especially in under-floor protection, engine decorative cover for passenger cars, wheel housing and inner door panel in automobiles.

Umberto Berardi et al. (2012), explained about the acoustic properties of some natural materials, this paper reports the acoustical characterization of the following natural fibers: kenaf, wood, hemp, coconut, cork, cane, cardboard, and sheep wool. The absorption coefficient and the flow resistance for samples of different thickness have been measured. By using existing theoretical models, this study also compares the measured behavior with the theoretically predicted behavior. This comparison shows the limits of theoretical models originally defined for porous materials with homogeneous fibers, when they are applied to natural materials. Finally, some suggestions for use of these natural fibers for sound absorption applications in buildings are reported.

Wenbin Zhu et al. (2015) reported about the effect of different fiber blend ratios and bulk densities at similar thicknesses on air permeability and sound absorption coefficient was investigated. The raw materials used were cotton, polyester, and bicomponent fibers to make acoustic nonwovens through the air-laid and thermal bonding processes. A uniform thermal-conductivity box was designed in order to make thermally bonded nonwovens with fixed thicknesses. The experimental results depict that the air flow resistance of three-layer nonwoven was 0.565 kPa·s/m, which was about four times greater than one-layer of 0.12 kPa·s/m. Sound absorption coefficient of 20% polyester-60% cotton-20% bicomponent nonwoven with lower bulk density was greater than

the 60% polyester-20% cotton-20% bicomponent nonwoven. The sound absorption coefficient varied based on which fabric side faced the testing apparatus.

Emedya Murniwaty Samsudin et al. (2017) reported about the utilization of Empty Fruit Bunch (EFB) and Mesocarp Fiber (MF) as the core materials in determination of significant relationship between three physical parameters (thickness, density and porosity) towards its sound absorption capabilities. The maximum NRC value for EFB & MF acoustic panel using 15% of UF as matrix binder achieved 0.58 at a density of 0.4g/cm^3 with a porosity value of 65.5%. This finding is considerable since the R^2 value for all regression lines represented by the curves exceeds the 98% reliability level. In conclusion, mixtures of EFB and MF in any forms (either dust or coir forms) has excellent potential to be used as raw fibers for acoustic absorbers replacing existing synthetic materials in the market.

Yasemin Korkmaz et al. (2012) discussed about the effects of physical parameters on sound absorption properties of nonwoven fabrics were investigated. Eight different nonwoven composites including different fiber types mixed with different ratios were tested. Along with sound absorption properties, thickness, weight per area, and air permeability parameters of the samples were measured. The increase in thickness and the decrease in air permeability results in an increase in sound absorption properties of the material. The samples including 70% cotton and 30% polyester resulted in the best sound absorption coefficient in the mid-to-high frequency ranges. The increase in the amount of fiber per unit area resulted in an increase in sound absorption of the material. Addition of acrylic and polypropylene into a cotton and polyester fiber mixture increased the sound absorption properties of the composite in the low and mid-frequency ranges.

Sanjay M R et al. (2015) discussed about a brief review has been carried out to make use of natural fibers (such as abaca, banana, bamboo, cotton, coir, hemp, jute, pineapple, sisal etc.) abundantly available in India. Glass Fiber Reinforced Polymers are mixing with natural fibers to increase Engineering and Technology applications. This paper presents a review on the mechanical properties of natural-glass fiber reinforced polymer composites. The mechanical and physical properties of natural fibers have varied from fiber to fiber. Natural fiber reinforced composites are used in many engineering applications, because of its superior properties such as specific strength, low weight, low cost, fairly good mechanical properties, non-abrasive, eco-friendly and bio-degradable characteristics. Incorporation of natural fibers with GFRP can improve the properties and used as an alternate material for glass fiber reinforced polymer composites.

Girish M N (2018) explained about the investigation of the application of Pineapple Leaf fiber reinforced Vinyl ester-based hybrid composites in acoustic barrier when the condition is worn driven. The obtained results are compared with a similar set of glass fiber reinforced Vinyl ester-based hybrid composites. It is evident that the values of Sound Pressure level (SPL) obtained in case of a Single cylinder 4S petrol engine, Vertical 4-S Diesel Engine and Air Compressor shows that the reduction in SPL can be achieved in its vicinity with maximum reduction at 1.5m from the source and a minimum reduction at 0.5m from the source for Pineapple leaf fiber (PALF) - Vinyl ester composites, thus after conducting the study we can say that the composites used can be recommended as a noise barrier also, and if the condition is erosion driven the composite material can sustain its property of reducing the Sound Pressure Level (SPL) dominantly and thus avoiding the sound levels.

Khai Hee OR et al. (2016) discussed the utilization of bres from the oil palm empty fruit bunch (OPEFB) to be an alternative natural acoustic material. The study was carried out by fabricating samples from raw OPEFB bres with different densities and thicknesses to observe their effects on the sound absorption performance. It has been demonstrated that the sound absorption performance can be improved by increasing the thickness of the sample and also by having optimum densities of bres. In particular for lower frequencies, this can be achieved by introducing air cavity gap behind the bre samples. Measurement of the normal incidence absorption coefficient in an impedance tube based on ISO 10534-2 found that the OPEFB bres can have absorption coefficient of 0.9 on average above 1 kHz. The sound absorption performance of OPEFB bres is also shown to be comparable to that of the commercial synthetic rock wools.

R.T. Durai Prabhakaran (2012) investigated a critical review of various materials for future for wind turbine blade design. In this study various fiber reinforcements, thermoset composites, thermoplastic composites, natural fiber composites, and hybrid composites were included and also explained disadvantages and disadvantages of various materials and its limitations that can give good insight for material selection for both large and small turbine blades. Finally, concluded the end-of-life of the wind turbine blades, thermoplastics blades can be recycled, which is a major advantage compared to its thermoset counterpart but it needs few more advances before implementing as blade materials. Whereas the natural fiber composites cannot be an immediate solution for large blades, but for smaller blades these materials can become a good choice. Furthermore, concluded that, another interesting material in future blades design could be hybrid composites.

Rahman et al. (2016) studied the surface treatment of the sisal fiber and its mechanical properties. Fiber surface modifications by mercerization were done before grafting with a view to improve the physic mechanical performance of sisal fibers. The effects of mercerization on shrinkage and fiber weight losses were monitored at different temperature and alkali concentration. They observed that, fiber shrinkage is higher at low temperature and 20% alkali treated sisal fibers yielded maximum shrinkage and weight losses. It was found that higher shrinkage of the polymer grafted fiber showed enhanced physic-mechanical properties. The grafting of alkali treated fiber shows an increase of polymer loading (about 56% higher) and tensile strength (about 27%) than 50% EMA grafted fiber. The fiber surface topology and the tensile fracture surfaces were characterized by scanning electron microscopy and were found improved interfacial bonding to the modified fiber–matrix interface.

Ersoy et al. (2015) investigated the sound absorption of an industrial waste, developed during the processing of tea leaves. Three different layers of tea-leaf-fiber waste materials with and without backing provided by a single layer of woven textile cloth were tested for their sound absorption properties. The experimental data indicate that a 1 cm thick tea-leaf-fiber waste material with backing provides sound absorption which is almost equivalent to that provided by six layers of woven textile cloth. Twenty millimeters thick layers of rigidly backed tea-leaf-fibers and non-woven fiber materials exhibit almost equivalent sound absorption in the frequency range between 500 and 3200 Hz.

P.N.B. Reis et al. (2009) Studied the flexural behavior of hand manufactured hybrid laminated composites with a sisal natural fiber/polypropylene core and two glass fibers/polypropylene surface layers at each side of the specimen. Laminate composites (LC) present an ultimate strength about 4% higher than the hybrid laminated composites (HLC) associated to changes in failure mechanisms, while the stiffness modulus was

also about 3.8% higher. Fatigue strength of hybrid laminated composites 21 is also about 20% lower than the laminated composites as consequence of the change of the failure mechanisms and of the different static strengths.

2.1 Summary of Literature Review

From the literature review the following points are arrived,

- By Increasing the fiber layers showed a positive improvement towards sound absorption coefficient.
- According to the amount of composite material added to the composition the better physical, mechanical and thermal properties can be obtained.
- The effect of the fiber sizes of micro porous materials distinctly promotes acoustic absorption with decreasing fiber size.
- The absorption properties of natural fibers can also be increased by increasing the bulk density of the material.
- Increasing air gap thickness usually shifts the peak of absorption coefficient towards lower frequency.
- The sound absorption property of natural fibers was superior to synthetic fibers such as glass and carbon fibers due to their unique hollow and multi scale structures.
- The acoustic behavior of natural fiber composites as high-quality absorbers, in combination with bio based granular materials.
- The effect of the mercerization or alkaline treatment will improve acoustic absorption performance.
- For large grains, the absorption is generally low due to low flow resistivity, but for smaller grains the absorption increases due to high flow resistivity and tortuosity.

2.2 Objectives

- To develop sustainable and green materials in order to have good acoustical environment to meet the stringent legislation and also to satisfy the customers.
- To develop a material having a less effect on occupational health and having better mechanical strength.
- To prove, industrial tea waste hybridized with pineapple/GFRP having better sound absorption co-efficient.
- Study the surface morphology using SEM analysis.

CHAPTER III

METHODOLOGY

A methodology of this project can be explained in the flowchart as shown in Fig 3.1.

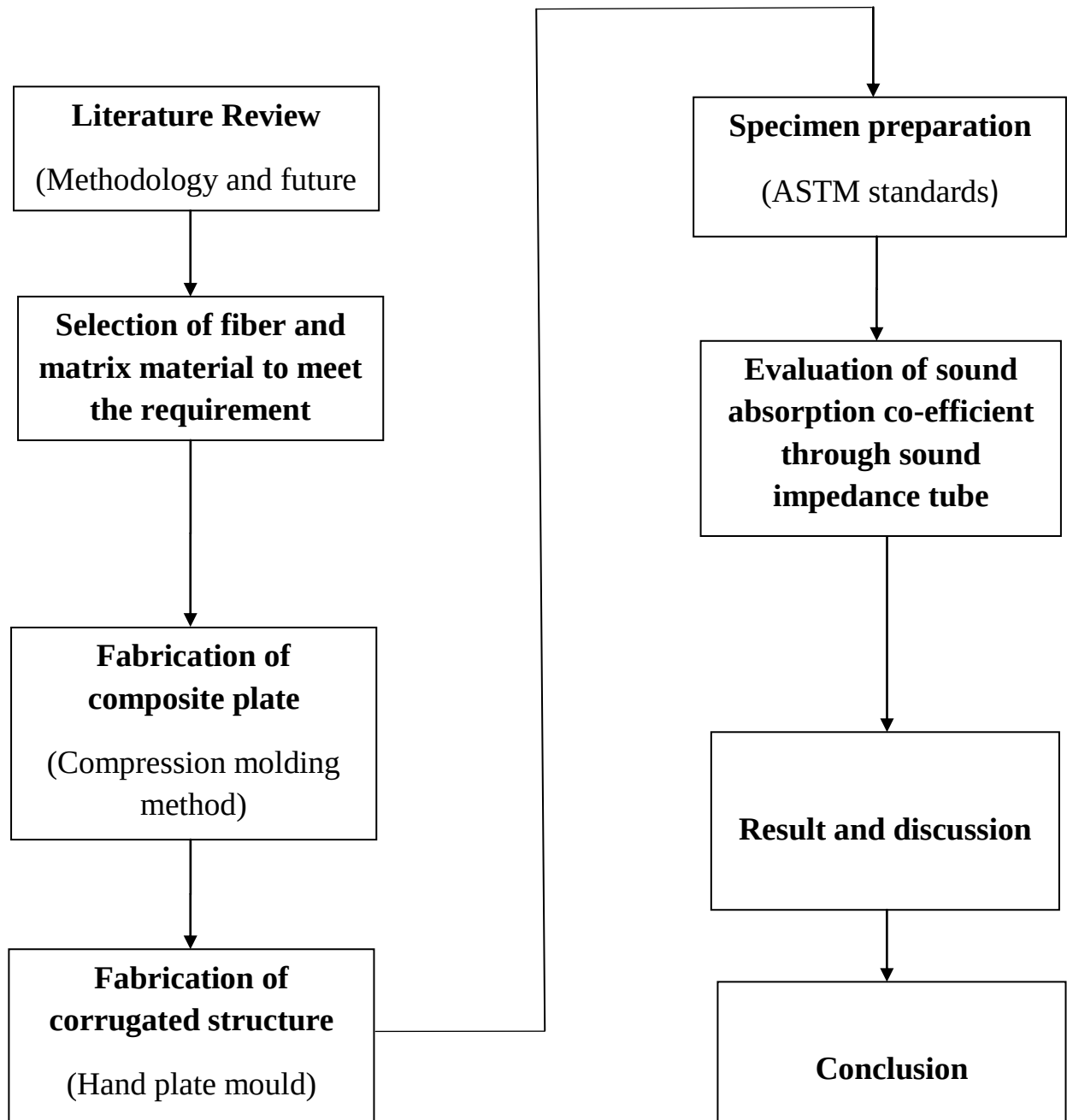


Fig 3.1 Flow chart for Methodology

3.1 Literature Review:

The study of various journals and papers were made related to the project. In which the literatures were collected for the references, previous patterns, structures, various natural fibers were already made with the acoustic journal and different types of structures and various parameters, such as physical parameters, mechanical parameters were studied. From the literatures the process and the design, natural fibers were finalized.

3.2 Material Requirement:

The materials required for the corrugated sandwich panels are pineapple natural fiber, tea powder, glass synthetic fiber. These are the basic materials other than that corrugated structured mould, flat laminate mould and pasting material araldite, PLA. These are the materials to meet the requirement and to manufacture the sandwich acoustic panel.

3.3 Fabrication of Flat Composite Plate:

The compression molding is a forming process in which the pineapple, tea and glass fibers layers are directly placed into a heated metal mould then it is softened by heat and therefore forced to conform the shape of mould, as the mould closes. Typically, compression molding machine open along a vertical axis. Initially upper and lower coils of the compression molding machine is set at 190c. Once the temperature has been reached, die filled with PLA & filler or pure PLA is placed in between two guided plates. The guided plates are placed in between two coils. Allow the setup for pre-heating for about 5 minutes. Now the die has been compressed at the pressure of about 100 bars, for about 25 minutes. Then allow the setup to get cooled for few minutes. Now remove the compression molded piece from the die.

3.4 Fabrication of Corrugated Structure:

The corrugated structure hand plate moulds are prepared to fabricate the structure. Corrugated structured mould is prepared from the stainless-steel asbestos sheet. It is perfect corrugated mould, easy readymade corrugated structured mould and the hand plate mould.

3.5 Sample Preparation:

The compression molded sandwich panels are cut into the circular shape according to the ASTM dimension standards. The cutting shape of the sound absorption co-efficient testing is in circular shape. This cutting piece is used for sound absorption co-efficient testing.

3.6 testing and Result:

The sound absorption co-efficient testing is a fundamental material science and engineering test in which the sample is subjected to control the sound or noise. This testing is performed through the sound impedance tube and the result is obtained in the form of graphical representation. Hence using the natural fibers in corrugated sandwich panels are undergone for testing and results are determined. The graphical representation results the sound absorption co-efficient for various frequencies and the maximum frequency of sound absorption at maximum level.

CHAPTER IV

PROBLEM STATEMENT AND OBJECTIVES

4.1 Problem statement

- Noise is one of the major air pollutants, which will have severe impact on living organisms. It is believed that acoustically treated industries not only improve the productivity of the workers but also provides good occupational safety.
- Majority of sound absorption materials are synthetic fibers and have poor mechanical properties. The usage of these materials will also severely affect the health and the environment.
- These synthetic acoustical materials are highly cost and used in advanced applications such as internal parts of automotive (door and window panels, hat shelves, and roofing) and building structures.
- Replacement of sustainable materials made from either natural or recycled materials will reduce the carbon footprint.
- Usually flat laminate structured acoustic panel contains more weight.

4.2 Objectives

- To develop sustainable and green materials in order to have good acoustical environment to meet the stringent legislation and also to satisfy the customers.
- To develop a material having a less effect on occupational health and having better mechanical strength.
- Fabrication of a new class of epoxy-based hybrid composites reinforced corrugated structure with bidirectional oriented short glass fibers, pineapple fibers and waste tea flakes to achieve requirements of good sound absorption capacity.

- To prove, industrial tea waste hybridized with pineapple/GFRP having better sound absorption co-efficient.
- Study the surface morphology using SEM analysis.

CHAPTER V

FABRICATION OF HYBRID COMPOSITES

5.1 Selection of natural fibers

(a) Waste tea powder

The waste green tea residues are especially attractive to be used as sound absorber. It is also feasible to achieve required sound absorption by adjusting tea residues thickness and combination structures with different facing fabrics. The waste tea powder is collected from a tea factory and its fiber are extracted by sieving it in a sieving machine in order to remove the fine dust particles are shown in Fig 5.1. A vibration made on machine sieves tea dust into 400 microns. As a result, external dust layers of the tea are removed.

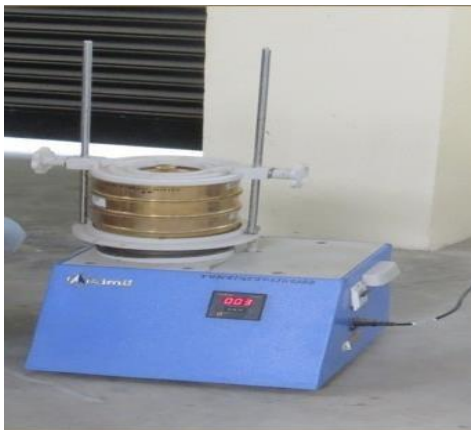


Fig 5.1 Sieve machine



Fig 5.2 Tea powder

(b) Pineapple fiber

Pineapple fiber mat are selected due to its high strength properties. Fiber mat is in Bi-directional structure, already treated with chemical methods and it is available on market. A Fig 5.2 shows dry pineapple fiber mat.



Fig 5.3 Pineapple fiber mate

(c) Glass fiber

Glass fiber has roughly comparable mechanical properties to other fibers such as polymers and carbon fiber. It is much cheaper and significantly less brittle when used in composites. Glass fibers are therefore used as a reinforcing agent for many polymer products to form a very strong and relatively lightweight fiber-reinforced polymer (FRP) composite material called glass-reinforced plastic (GRP), also popularly known as "fiberglass".



Fig 5.4 Woven glass fiber

E-glass Fiber (GSM 200) – aluminoborosilicate glass with less than 1% weight alkali oxides, mainly used for glass-reinforced plastics is selected as one of the fiber materials in the form of woven fabric as shown in Fig 5.3.

5.2 Selection of matrix material

5.2.1 Epoxy Resin

Epoxy resins, also known as poly-epoxides, are a class of reactive pre-polymers and polymers which contain epoxide groups. Epoxy resins may be reacted (cross-linked) either with themselves through catalytic homopolymerization, or with a wide range of co-reactants including poly-functional amines, acids (and acid anhydrides), phenols, alcohols and toils. These co-reactants are often referred to as hardeners or curatives, and the cross-linking reaction is commonly referred to as curing shown in Fig 5.2.1 Reaction of poly-epoxides with themselves or with poly-functional hardeners forms a thermosetting polymer, often with high mechanical properties, temperature and chemical resistance. Epoxy has a wide range of applications, including metal coatings, use in electronics / electrical components/LED, high tension electrical insulators, fiber-reinforced plastic materials and structural adhesives

Epoxy resins are low molecular weight pre-polymers or higher molecular weight polymers which normally contain at least two epoxide groups. The epoxide group is also sometimes referred to as a glycidyl or oxidant group.

A wide range of epoxy resins are produced industrially. The raw materials for epoxy resin production are today largely petroleum derived; although some plant derived sources are now becoming commercially available (e.g. Plant derived glycerol used to make epichlorohydrin).

Epoxy adhesives are a major part of the class of adhesives called "structural adhesives" or "engineering adhesives" that includes polyurethane, acrylic, cyanoacrylate and other chemistries.

These high-performance adhesives are used in the construction of aircraft, automobiles, bicycles, boats, golf clubs, skis, snowboards, and other applications where high strength bonds are required. Epoxy adhesives can be

developed to suit almost any application. Epoxy adhesives are better in heat and chemical resistance than other common adhesives.

A polyvinyl Alcohol (PVA) acts as a best releasing agent to remove a composite plate from cavity die without any ease. Usually the resin and hardener are mixed with a constant mixing ratio of 5:1 e.g. for 1kg of resin 200 grams of hardener is to be mixed and properly stirred for graft copolymerization

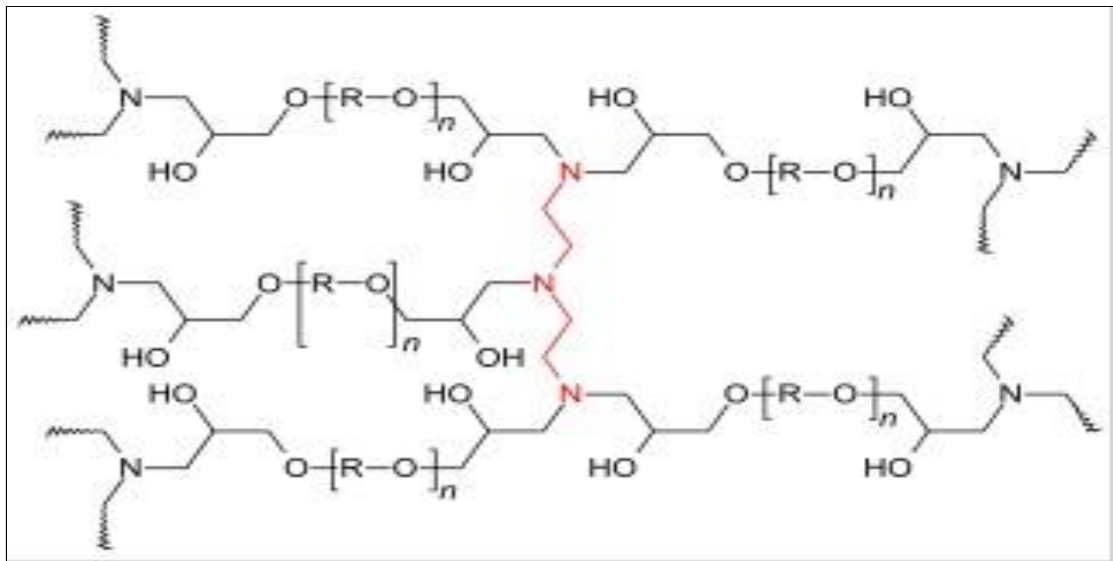


Fig 5.5 Structure of cured epoxy glue

5.2.2 Advantages of epoxy resin

- Better adhesive properties (the ability to bond to the reinforcement or core)
- Superior mechanical properties (particularly strength and stiffness)
- Improved resistance to fatigue and micro cracking
- Reduced degradation from water ingress (diminution of properties due to water penetration)
- Increased resistance to osmosis (surface degradation due to water permeability)

5.2.3 Properties of Araldite LY 556 / Hardener HY 917

Araldite LY 556		
Aspect (visual)	clear, pale yellow liquid	
Colour (Gardner, ISO 4630)	≤ 2	
Epoxy content (ISO 3000)	5.30 - 5.45	[eq/kg]
Viscosity at 25 °C (ISO 9371B)	10000 - 12000	[mPa s]
Density at 25 °C (ISO 1675)	1.15 - 1.20	[g/cm ³]
Flash point (ISO 2719)	> 200	[°C]
Storage temperature (see expiry date on original container)	2 - 40 °C	[°C]
Hardener HY 917		
Aspect (visual)	clear liquid	
Colour (Gardner, ISO 4630)	≤ 2	
Viscosity at 25 °C (ISO 9371B)	50 - 100	[mPa s]
Density at 25 °C (ISO 1675)	1.20 - 1.25	[g/cm ³]
Flash point (ISO 2719)	195	[°C]
Storage temperature	2 - 40 °C	[°C]

Fig 5.6 Properties of araldite LY 556 and hardener HY 917

5.2.4 Alkalization of tea fibers

Tea dust fibers were soaked in 5% weight of NaOH solution at normal room temperature for 3 days, maintaining a liquor ratio of 10:1. The fibers were washed several times with water to remove any alkali solution sticking to the fibers surface, neutralized with dilute acetic acid and then washed again with water. Finally, the result fibers were dried at 60°C for 48 hours and also dried off with natural sunlight for 8 hours before the examination. The chemical treatment of tea is shown in Fig 5.6



(a) Soaking of fibers in NaOH

(b) Manual drying

Fig 5.7 Chemical treatment of tea fiber (Alkalization)

A Fig 5.7 shows that decreases in the structural compositions between the treated and untreated fibers.



Fig 5.8 Untreated and treated tea fiber.

5.2.5 Rule of mixture

Kelly and Tyson (1965) first proposed the rule of mixtures model for continuous fibers with uniaxial orientation. The rule of mixtures is commonly used to estimate the modulus and strength. In the fiber direction, assuming perfect bonding between fiber and matrix, the properties are,

$$E_c = E_f V_f + E_m V_m$$

$$\sigma_c = \sigma_f V_f + \sigma_m V_m$$

$$\text{Where, } V_m = 1 - V_f$$

Since E_c and σ_c are the modulus and strength of composites in the fiber direction. E_m and E_f are the modulus of matrix and fiber. V_m and V_f are the

volume fractions of matrix and fiber. σ_f and σ_m are the stresses in fiber and matrix respectively, at fiber failure.

5.2.6 Designation of composite plates

The designations of composite laminate used for the experiment is tabulated in Table 5.1 with volume fraction of matrix material as 60wt% and remaining 40%wt was split up with different fractions of, glass and tea fibers.

Table 5.1 Designation of composite laminate

S. No	Glass fibers		Pineapple fibers		Waste tea fibers		Epoxy resin	
	(%)	(grams)	(%)	(grams)	(%)	(grams)	(%)	(grams)
STGE 1	10 %	60 grams	10 %	60 grams	20%	120 grams	60%	360 grams
STGE 2	10 %	60 grams	20 %	120 grams	10%	60 grams	60%	360 grams
STGE 3	10 %	60 grams	15 %	90 grams	15%	90 grams	60%	360 grams
STGE 4	10 %	60 grams	5%	30 grams	25%	150 grams	60%	360 grams
STGE 5	10 %	60 grams	25 %	150 grams	5%	30 grams	60%	360 grams

***PTGE – Pineapple/Tea/Glass/Epoxy composites**

5.3 Fabrication of composite plate

In compression molding, the first step is to stack up the fibers with stacking sequences as shown in Fig 5.8 by hand lay-up method. Then it is

compressed under the temperature and pressure maintained over a period of time.

In Fig 5.8 the layers of fiber are supposed to be made as a composite plate is shown. The outer layers are waste tea powder and the inner layers are glass fiber and in between the tea and glass fiber pineapple fiber are placed.

Tea
Pineapple
Glass
Glass
Pineapple
Tea

Fig 5.9 Stacking sequence of composite.

5.4 Principle of compression molding

Compression molding is a process of molding in which a feeding material is placed into an open, heated mold cavity. The mold is then closed with a top plug and compressed with large hydraulic presses in order to have the material contact all areas of the mold. The charge cures in the heated mold.

5.4.1 Compression molding process

The Flowchart of compression molding process is shown in Fig 5.9 The temperature is maintained at 130°C and pressure at 35 kg/cm² for 3 hours. For first two and a half hours both temperature and pressure are maintained and for the remaining half an hour only pressure is maintained.

The flat laminate plate manufactured using the compression molding process. The process carried out with the constant temperature and the pressure

for the specific period of time.

The flat laminate plate has a different mold with different measurements, in which based on our requirement the measurement of 27cm×27cm and thickness of 2mm. It is the standard measurements obtained from ASTM.

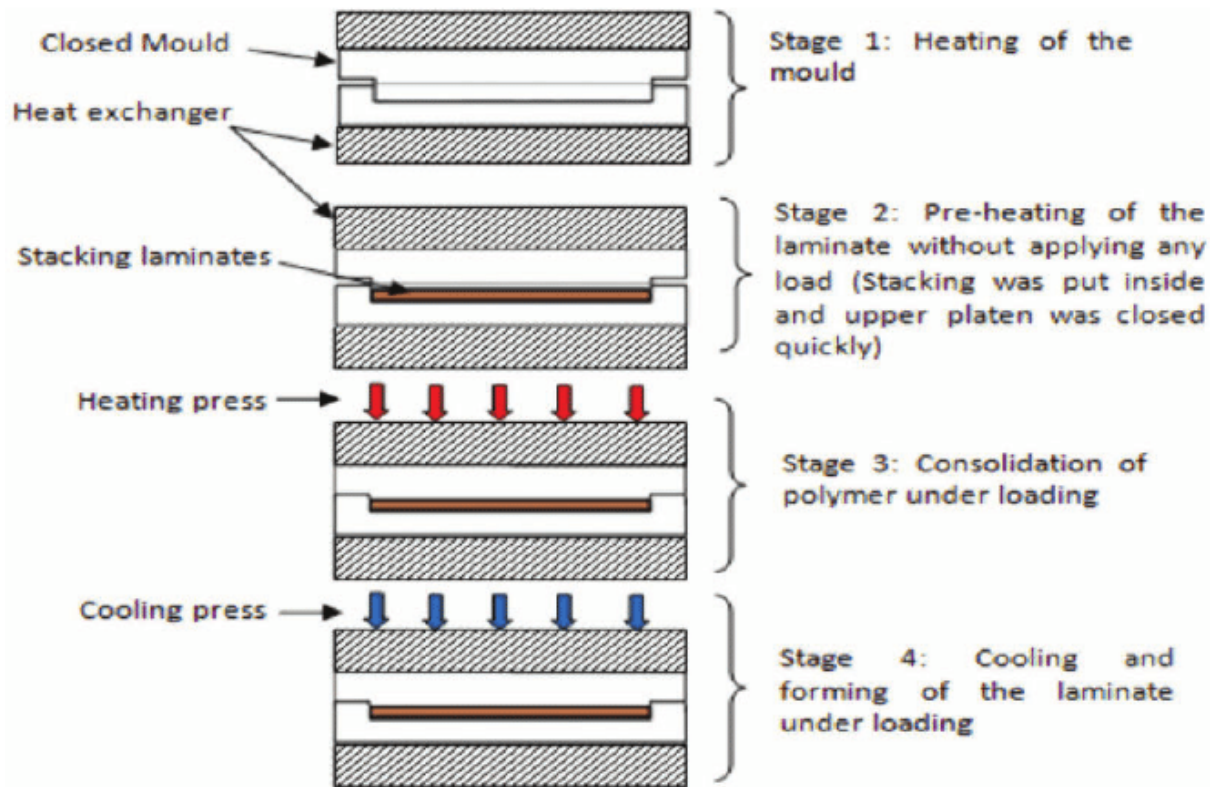


Fig 5.10 compression molding process

The composite plates are listed in the Table 5.1. These plates are then marked as per ASTM standards for each test and squeezed out.

The following are the four stages involved in compression process as shown in Fig 5.9,

1. Heating of mold
2. Pre-heating the laminate
3. Applying load
4. Cooling

According to the step by step process the hybrid laminates are prepared. As

Shown in the Fig 5.10 the mixer is mixed and used to make the composite materials.

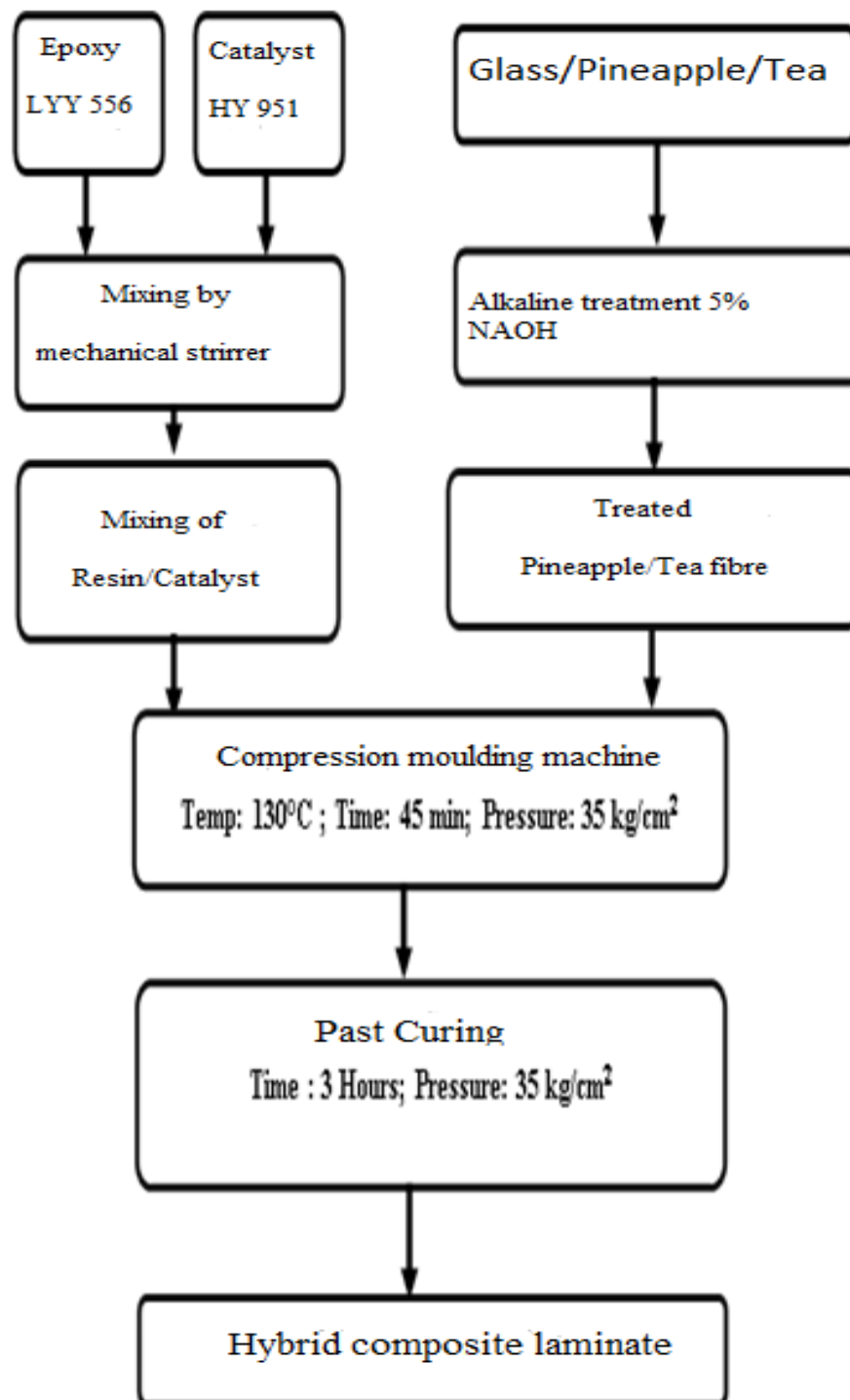


Fig 5.11 Process of making hybrid composite



Fig 5.12 Compression molding machine.

5.4.2 Hybrid composite mould

Injection mold construction is the process of creating molds that are used to perform injection molding operations using an injection molding machine. These are generally used to produce plastic parts using a core and a cavity. Molds are designed as two-plate or three-plate molds, depending on the type.

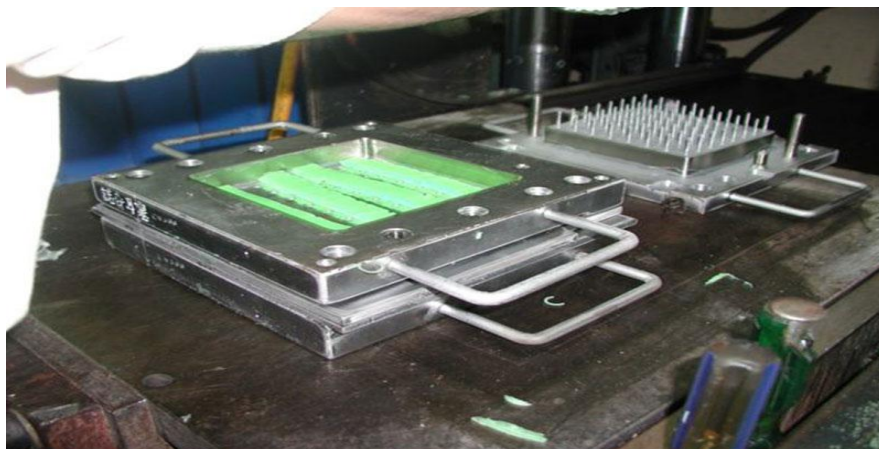


Fig 5.13 Mould plate

5.5 Sound absorption test specimen

The sound absorption specimens were prepared in accordance with ASTM E1050-10 of dimensions 99.5 cm diameter (lower frequency tube) and 29.5 cm diameter (higher frequency tube) are listed in Table 5.2 The core layer of glass and sisal symmetric stack up is filled with tea fibers for requirement of sound insulation.

5.5.1 Test specimen cutting

The specimen is prepared by using the water jet cutting process. In water jet cutting process the first step includes the coding of the required shape and dimensions to be entered in the computer, next step the specimen is to be placed in the cutting machine with the perpendicular of cutting nozzle.

Once the water jet cutting machine is on the nozzle starts cutting for the code entered in the computer for the required dimensions and then finally the test specimen is obtained.



Fig 5.14 Water jet cutting machine

Table 5.2 Different specimens for sound absorption test

Composition	Volume fraction of tea fibers (%)	Sound absorption test specimen
Tea/Epoxy	60%	
Tea/Epoxy	70%	
Glass/Tea/Epoxy	20%	
Pineapple/Glass/Tea/Epoxy	15%	

CHAPTER VI

EXPERIMENTAL ANALYSIS

6.1 Sound impedance test

Acoustic absorption refers to the process by which a material, structure, or object takes in sound energy when sound waves are encountered, as opposed to reflecting the energy. Part of the absorbed energy is transformed into heat and part is transmitted through the absorbing body. When sound from a loudspeaker collides with the walls of a room part of the sound's energy is reflected, part is transmitted, and part is absorbed into the walls. As the waves travel through the wall they deform the material thereof (just like they deformed the air before). This deformation causes mechanical losses via conversion of part of the sound energy into heat, resulting in acoustic attenuation, mostly due to the wall's viscosity.



Fig 6.1 Sound impedance test tube

Similar attenuation mechanisms apply for the air and any other medium through which sound travels. The fraction of sound absorbed is governed by the acoustic impedances of both media and is a function of frequency and the incident angle. Size and shape can influence the sound wave's behavior if they interact with its wavelength, giving rise to wave phenomena such as standing waves and diffraction. The Acoustic testing equipment is shown in Fig 6.1.

The variation in sound absorption co-efficient for different combinations is listed in Table 6.1

Table 6.1 Sound absorption co-efficient

Composition	Volume of tea fiber	Frequencies (Hz)							NRC
		250 Hz	500 Hz	1000 Hz	2000 Hz	2500 Hz	5000 Hz	6300 Hz	
Tea/Epoxy	60%	0.3	0.4	0.12	0.23	0.34	0.59	0.37	0.335
Tea/Epoxy	70%	0.17	0.25	0.33	0.35	0.39	0.81	0.57	0.41
Glass/Tea/Epoxy	10%	0.03	0.04	0.09	0.17	0.32	0.6	0.51	0.251
pineapple/Glass/Tea/Epoxy	20%	0.0503	0.0785	0.0588	0.4121	0.5019	0.6489	0.5952	0.332

*- NRC – Noise Reduction Co-efficient

6.2 Experimental setup

The measurement of sound absorption coefficient was conducted by using the impedance tube testing according to ISO 10534-2 (European Committee for Standardization, 2001). The measurement setup is shown in Figure 5. The sample of GTPF was fitted tightly into the 33 mm diameter sample holder (Figure 6) which located at one end of the impedance tube. The circular surface of the sample was faced normal to the direction of incoming sound waves. The other end of the impedance tube was the location of a loudspeaker.

A white noise was generated from the computer into the impedance tube through the loud-speaker. The loudspeaker was operated for 10 minutes before the measurement was taken to allow the temperature inside the impedance tube to stabilize. Two 1/2 pre-polarized free field microphones (GRAS 40AE) with 1/2 CCP pre-amplifier (GRAS 26CA) were located between the sample and the loudspeaker to record the built-up sound pressure inside the impedance tube.

The sensitivity of microphones was calibrated by using Brüel & Kjær sound calibrator type 4231 at 114 dB level and 1 kHz. The RT Pro Photon+ analyzer was used as the data acquisition system to process the recorded signals and to obtain the transfer function between the two microphones.

MATLAB software was used to calculate the sound absorption coefficient as a function of frequency. In this experiment, the valid frequency is between 500 Hz to 4.5 kHz due to the size of the microphones and the tube diameter.

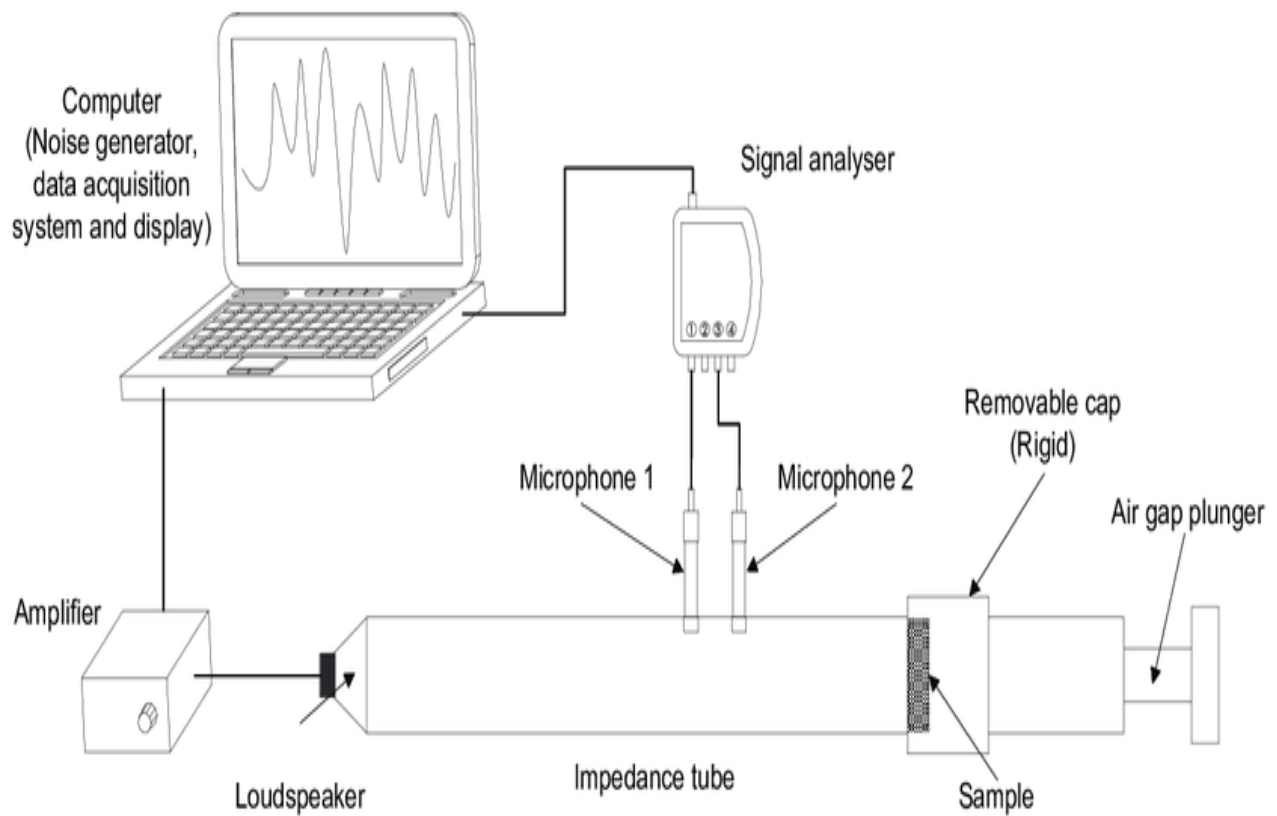


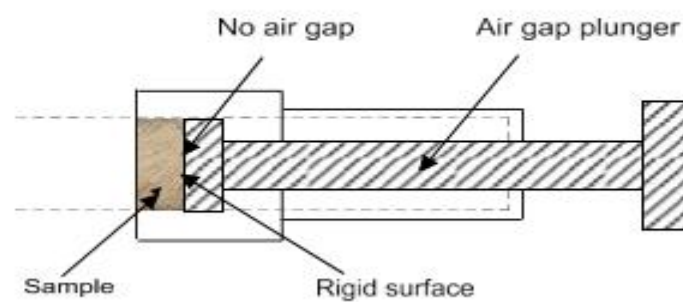
Fig 6.2 The experimental setup of impedance tube testing according to ISO 10534-2.



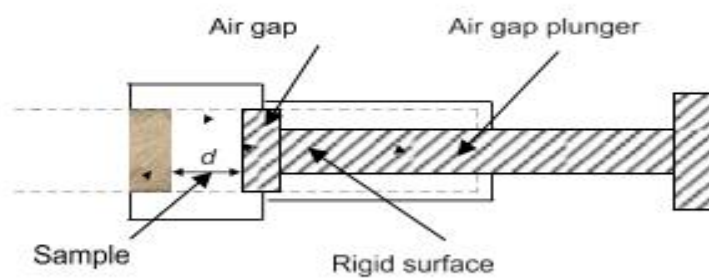
Fig 6.3 Sample holder of an impedance tube

The PALF sample was fitted into the sample holder of an impedance tube for the measurement of sound absorption coefficient: (a) an empty sample holder and (b) PALF sample was fitted tightly into the sample holder.

The impedance tube also consists of a mobile air gap plunger, which allows the application of air gap layer behind the sample. The illustration of the air gap layer, denoted as d is shown in Figure 6.4.



(a) No air gap ($d = 0$ mm)



(b) With air gap layer.

Fig 6.4 The illustration of air gap layer behind the sample

CHAPTER VII

RESULTS AND DISCUSSIONS

7.1 Sound absorption test

The composites with 15 wt.% of pineapple fibers and 5 wt.% of tea fibers and 5 wt.% glass fibers were characterized by the relatively weak sound absorption shown in Fig 7.15 The values of sound absorption coefficient (α) varies between 0.3 – 0.91 by varying frequencies. However, with increasing fiber content, an increase in sound absorption coefficient can be observed, especially at higher concentrations.

The sound absorption coefficient generally increases with increase of frequency.

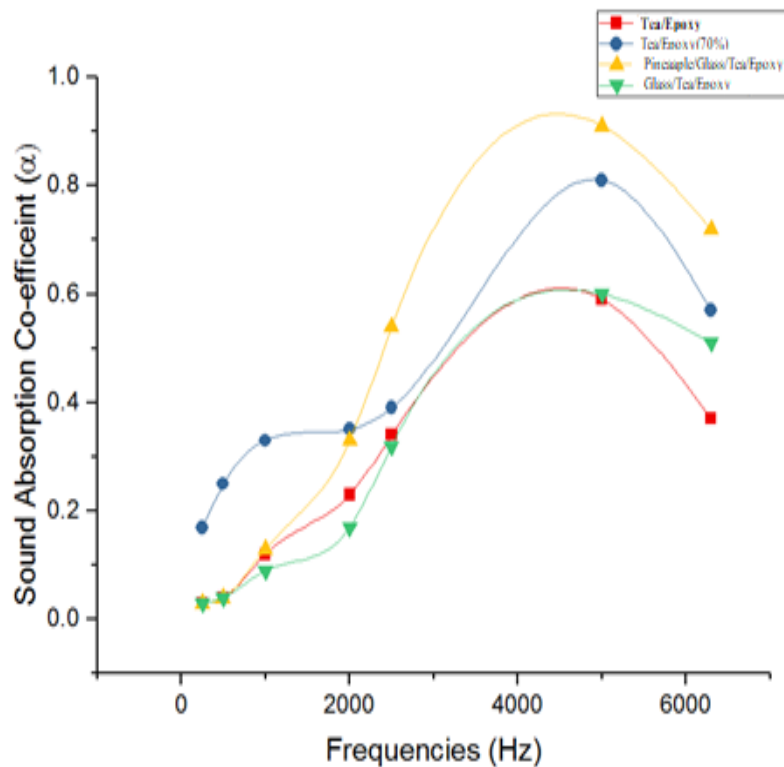


Fig 7.1 Sound absorption co-efficient of hybrid composites

The absorption coefficient of the hybrid composite is increases with the increase in frequency. The result shows that the absorption co-efficient of the tea, pineapple and glass are at is maximum at the frequency range of 4000-6000Hz.

It shows that the combination of glass and epoxy are at 0.6 and by adding natural fibers like tea powder and pineapple the sound absorption co-efficient will be increased up to the 0.91. In order to make a visual comparison among the different fibers, the noise reduction coefficients (NRCs), defined as the arithmetic mean value of sound absorption coefficient at 250, 500, 1000, 2000, 2500, 5000, 6000 Hz.

Table 7.1 Sound absorption test result

Sample	Volume fraction of pineapple and tea fiber	Frequencies(Hz)							SAC
		250	500	1000	2000	2500	5000	6000	
C1	20%	0.0503	0.0785	0.0588	0.4121	0.5019	0.9289	0.7952	0.332

At high frequencies, it gradually increased back and reached the peak absorption coefficient of 0.91 at 5000 Hz. It is still capable to be used as good absorber because the sound absorption coefficient range is in still high absorptive character. The sound absorption coefficients were measured with the ASTM E-10534, sound impedance test tube. The results of sound absorption coefficients in the frequency ranges from 250 to 6000 Hz are tabulated in the table.

CHAPTER VIII

CONCLUSIONS & FUTURE WORKS

From this project the sound absorption properties of pineapple-tea-glass fiber reinforced hybrid composites were evaluated by corrugated structure and different layer ratios of glass fiber, pineapple and tea fibers. The corrugated structure is mainly designed for to reduce weight and to introduce the different types of structure in acoustic panel using the natural fibers. The acoustic panel designed with two different structures, such as flat laminate plate structure and corrugated structure. These structures are arranged in the order of first and last to be flat laminate plate structure, in between the flat laminate plates corrugated structure is bonded, hence it is also called as sandwich panels. The sample is taken in the circle structure for particular diameter, using the water jet cutting process. The sample is tested through sound impedance test tube method. The sound absorption capacity of acoustic panel is obtained. The sound absorption capacity of the composites was found to be increasing with the hybridization of pineapple content with glass fiber and tea fiber.

- 20:10 weight percentage of pineapple and tea designation has maximum advantages than other designations.
- The sound absorption co-efficient of this hybrid composite also increased when compared with existing product.
- The sound absorption characteristic of the pineapple leaf fibers has been presented. From the measurement, the PALF sound absorber can have averaged absorption coefficient of 0.9 above 2 kHz for thickness 20 mm and of 0.8 above 1 kHz for thickness 30 mm for the same fiber density.
- The results suggest that PALF to be the next possible natural and sustainable sound absorber to replace the traditional synthetic sound absorbers.

These increases in properties of the studied material could make this applicable in components of automobiles like dashboards, seat bases, theaters, aircraft interior paneling, industrial applications, speaker compartment, sound proofing materials and furniture.

8.1 Future works

- The mechanical properties of the corrugated structured acoustic panel to be tested and analyzed.
- Thermal analysis of hybrid composites in order to test their glass transition temperature to fulfill the requirements for various applications.
- To compare these properties with other hybridization of natural fibers such as banana, jute and coir.
- Analysis for replacement of existing product with these hybrid composites.
- The physical parameters on natural fibers to be analyzed.

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LIST OF PUBLICATIONS

Patent Publication

- [1]. S.Gokulkumar, S. Reavanth, A. Suryaprakaash, & M. Vidur Varshan, “*Camellia Sinensis / Ananas Comosus Fiber Based Uni-Directional Corrugated Panel for Sound Damping Application*”, 2021, Ind. Patent No: 202141000670 A